MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
DECADE COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced Schottky TTL decade counter. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Synchronous 4-bit decade counter (asynchronous master reset)
02	Synchronous 4-bit decade counter (synchronous reset)
03	Synchronous 4-bit up/down decade counter (with mode control)
04	Synchronous 4-bit up/down decade counter (asynchronous master reset)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
(device types 01, 02, 03, 04)	303 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 4/:	
Low level	33 maximum
High level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Width of clock pulse, high ($\overline{PE}$ = high, low)	
Device types 01, 02	5.0 ns minimum
Width of clock pulse, low ($\overline{PE}$ = high)	
Device types 01, 02	8.0 ns minimum
Width of clock pulse, low ($\overline{PE}$ = low)	
Device types 01, 02	6.0 ns minimum
Width of master reset pulse, low	
Device type 01	7.0 ns minimum
Width of $\overline{PL}$ pulse, low	
Device type 03	8.5 ns minimum
Width of $\overline{PL}$ pulse, low	
Device type 04	7.5 ns minimum
Width of clock pulse, low	
Device type 03	7.0 ns minimum
Width of CPU or CPD pulse, low	
Device type 04	7.0 ns minimum
Width of master reset pulse, high	
Device type 04	6.0 ns minimum
Width of CPU or CPD pulse, low (change of direction)	
Device type 04	12.0 ns minimum
Setup time Pn high to clock pulse	
Device types 01, 02	5.5 ns minimum
Setup time Pn low to clock pulse	
Device types 01, 02	5.5 ns minimum
Setup time $\overline{PE}$ or $\overline{SR}$ high to clock pulse	
Device types 01, 02	13.5 ns minimum
Setup time $\overline{PE}$ or $\overline{SR}$ low to clock pulse	
Device types 01, 02	10.5 ns minimum
Setup time CEP or CET high to clock pulse	
Device types 01, 02	13.0 ns minimum
Setup time CEP or CET low to clock pulse	
Device types 01, 02	7.5 ns minimum
Setup time U/D high to clock pulse	
Device type 03	12.0 ns minimum
Setup time U/D low to clock pulse	
Device type 03	12.0 ns minimum
Setup time Pn high to $\overline{PL}$	
Device types 03, 04	6.0 ns minimum
Setup time Pn low to $\overline{PL}$	
Device types 03, 04	6.0 ns minimum
Setup time $\overline{CE}$ low to clock pulse	
Device type 03	10.5 ns minimum
Hold time Pn high to clock pulse	
Device types 01, 02	2.5 ns minimum
Hold time Pn low to clock pulse	
Device types 01, 02	2.5 ns minimum
Hold time $\overline{PE}$ or $\overline{SR}$ high to clock pulse	
Device types 01, 02	2.0 ns minimum
Hold time $\overline{PE}$ or $\overline{SR}$ low to clock pulse	
Device types 01, 02	0.0 ns minimum
Hold time CEP or CET high to clock pulse	
Device types 01, 02	0.0 ns minimum
Hold time CEP or CET low to clock pulse	
Device types 01, 02	0.0 ns minimum

4/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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Hold time Pn high to $\overline{PL}$	
Device types 03, 04	2.0 ns minimum
Hold time Pn low to $\overline{PL}$	
Device types 03, 04	2.0 ns minimum
Hold time $\overline{U}$/D high to clock pulse	
Device type 03	0.0 ns minimum
Hold time $\overline{U}$/D low to clock pulse	
Device type 03	0.0 ns minimum
Hold time $\overline{CE}$ low to clock pulse	
Device type 03	0.0 ns minimum
Recovery time master reset to clock pulse	
Device type 01	7.0 ns minimum
Recovery time $\overline{PL}$ to clock pulse	
Device type 03	7.5 ns minimum
Recovery time master reset to CPU or CPD	
Device type 04	4.5 ns minimum
Recovery time $\overline{PL}$ to CPU or CPD	
Device type 04	8.0 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

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3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High-level input current	I_{IH1} I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V V_{CC} = 5.5 V, V_{IH} = 7.0 V	All		20 100	µA µA
Low-level input current	I_{IL1} I_{IL2}	V_{CC} = 5.5 V, V_{IL} = 0.5 V V_{CC} = 5.5 V, V_{IL} = 0.5 V	All 01,02 03,04	-.03 -.50 -.75	-0.6 -1.2 -1.8	mA mA mA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V, V_{OS} = 0.0 V	All	-60	-150	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	All		55	mA
Maximum count frequency	f_{MAX}	V_{CC} = 5.0 V	All	70		MHz
Propagation delay time low to high level		V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 5				
CP to Qn	t_{PLH1}		03	3.0	9.5	ns
CP to Qn	t_{PHL1}		03	5.0	13.5	ns
CP $\overline{U}$/D to Qn	t_{PLH1}		04	4.0	10.0	ns
CP $\overline{U}$/D to Qn	t_{PHL1}		04	5.5	14.0	ns
CP to Qn $\overline{PE}$ = (high)	t_{PLH1}		01,02	3.5	9.0	ns

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
CP to Qn $\overline{PE}$ = (high)	t_{PHL1}		01,02	3.5	11.5	ns
CP to T_C	t_{PLH2}		03	6.0	16.5	ns
CP to T_C	t_{PHL2}		03	5.0	13.5	ns
CPU to $\overline{TCU}$	t_{PLH2}		04	4.0	10.5	ns
CPU to $\overline{TCU}$	t_{PHL2}		04	3.5	9.5	ns
$\overline{PL}$ to Qn	t_{PLH3}		04	5.0	13.5	ns
$\overline{PL}$ to Qn	t_{PHL3}		04	5.5	15.0	ns
CPD to $\overline{TCD}$	t_{PLH4}		04	4.0	10.5	ns
CPD to $\overline{TCD}$	t_{PHL4}		04	3.5	9.5	ns
CP to Qn $\overline{PE}$ = (low)	t_{PLH2}		01,02	3.0	10.0	ns
CP to Qn $\overline{PE}$ = (low)	t_{PHL2}		01,02	3.0	10.0	ns
CP to $\overline{RC}$	t_{PLH3}		03	3.0	9.5	ns
CP to $\overline{RC}$	t_{PHL3}		03	3.0	9.0	ns
CP to TC	t_{PLH3}		01,02	4.0	16.5	ns
CP to TC	t_{PHL3}		01,02	4.0	15.5	ns
PN to Qn	t_{PLH4}		03	3.0	9.0	ns
PN to Qn	t_{PHL4}		03	6.0	16.0	ns
PN to Qn	t_{PLH5}		04	3.0	8.5	ns
PN to Qn	t_{PHL5}		04	6.0	16.5	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
CET to TC	t_{PLH4}		01,02	2.5	9.0	ns
CET to TC	t_{PHL4}		01,02	2.5	9.0	ns
$\overline{CE}$ to $\overline{RC}$	t_{PLH5}		03	3.0	9.0	ns
$\overline{CE}$ to $\overline{RC}$	t_{PHL5}		03	3.0	9.0	ns
$\overline{MR}$ to Qn	t_{PHL5}		01	5.5	14.0	ns
$\overline{MR}$ to TC	t_{PHL6}		01	4.5	12.5	ns
$\overline{PL}$ to Qn	t_{PLH6}		03	5.0	13.0	ns
$\overline{PL}$ to Qn	t_{PHL6}		03	5.5	14.5	ns
$\overline{MR}$ to $\overline{TCU}$	t_{PLH6}		04	6.0	15.0	ns
[illegible] to $\overline{TCD}$	t_{PHL6}		04	7.0	16.0	ns
U/D to $\overline{RC}$	t_{PLH7}		03	7.0	22.0	ns
$\overline{U}$/D to $\overline{RC}$	t_{PHL7}		03	5.5	14.0	ns
$\overline{MR}$ to Qn	t_{PHL11}		04	6.5	16.0	ns
[illegible] TC	t_{PLH8}		03	4.0	13.5	ns
[illegible] TC	t_{PHL8}		03	4.0	12.5	ns
[illegible] to $\overline{TCU}$	t_{PLH7}		04	7.0	18.5	ns
[illegible] to $\overline{TCU}$	t_{PHL7}		04	7.0	17.5	ns
[illegible] $\overline{TCD}$	t_{PLH8}		04	7.0	18.5	ns
$\overline{PL}$ to $\overline{TCD}$	t_{PHL8}		04	7.0	17.5	ns
[illegible] to $\overline{TCU}$	t_{PLH9}		04	7.0	16.5	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Pn to $\overline{TCU}$	t_{PHL9}		04	6.5	16.5	ns
PN to $\overline{TCD}$	t_{PLH10}		04	7.0	16.5	ns
PN to $\overline{TCD}$	t_{PHL10}		04	6.5	16.5	ns

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Subgroups (see table III) Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

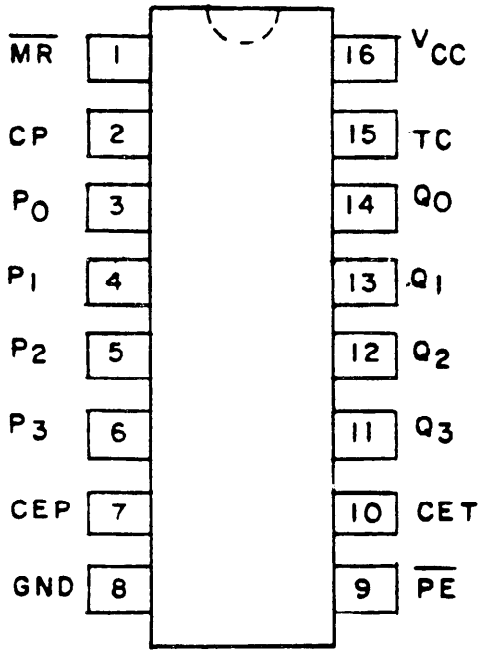
5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

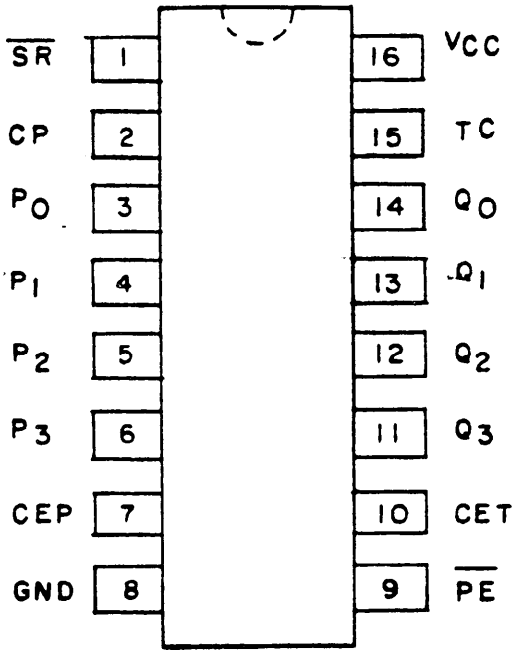
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Cases E and F

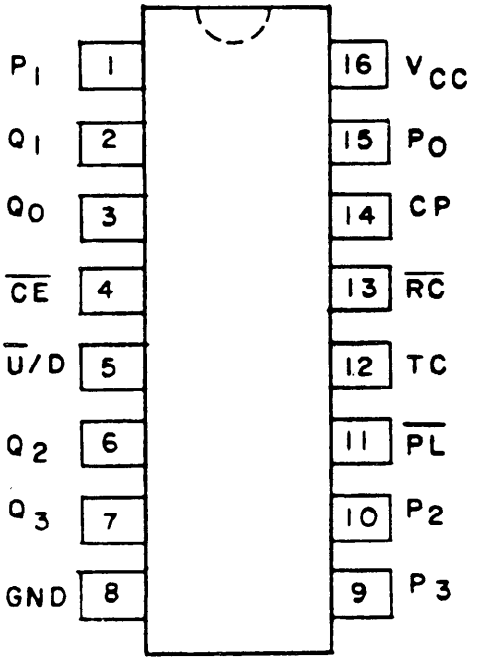
Device type 01



Device type 02



Device type 03



Device type 04

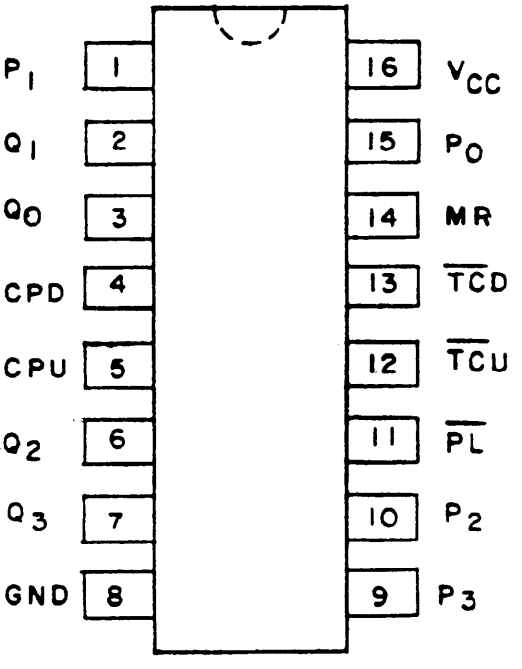
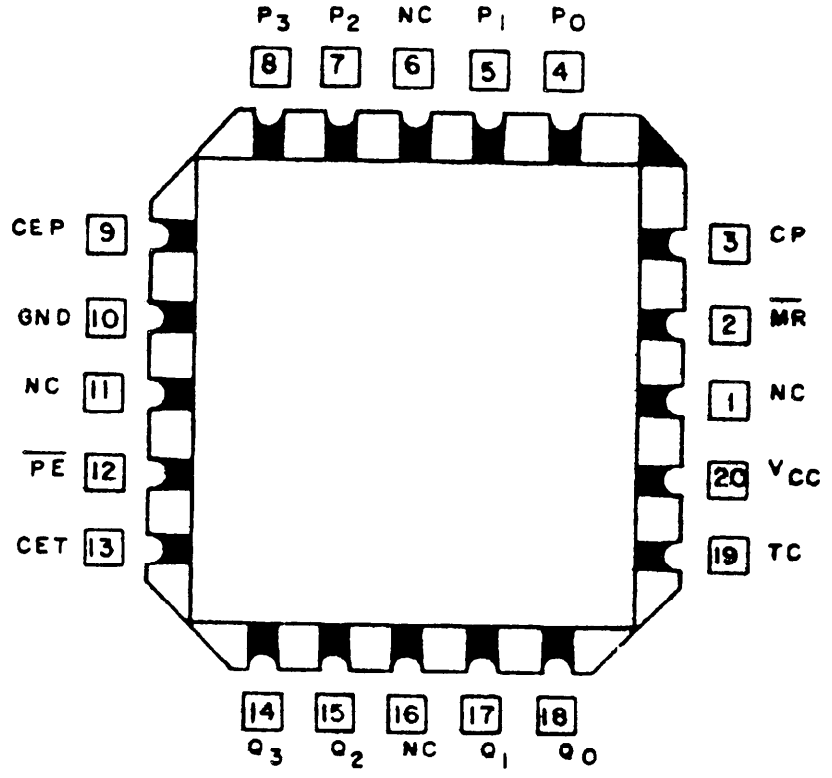


FIGURE 1. Terminal connections (top view).

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Case ?

Device type 01



Device type 02

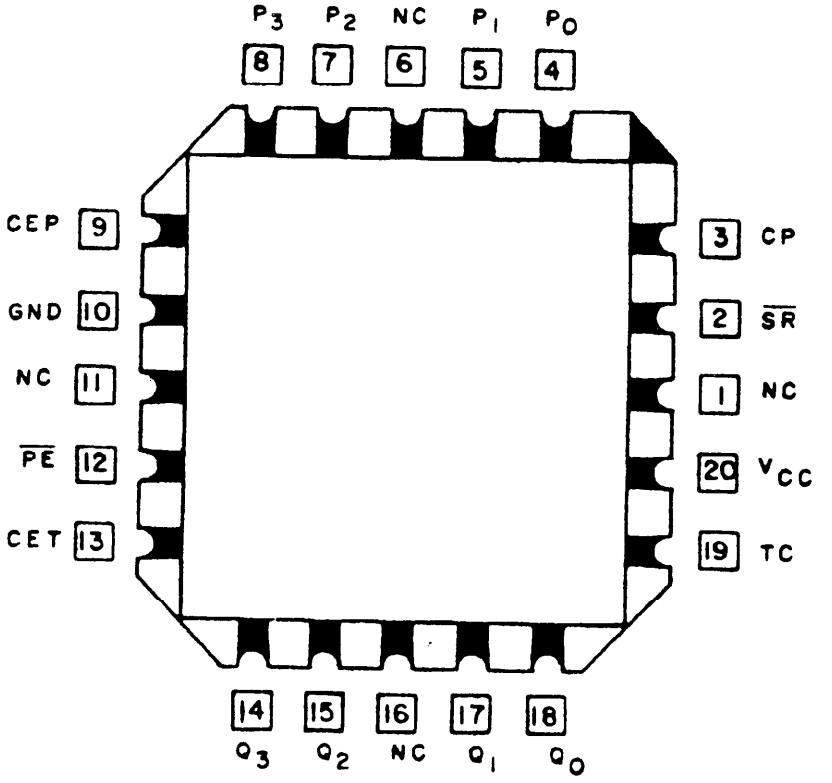
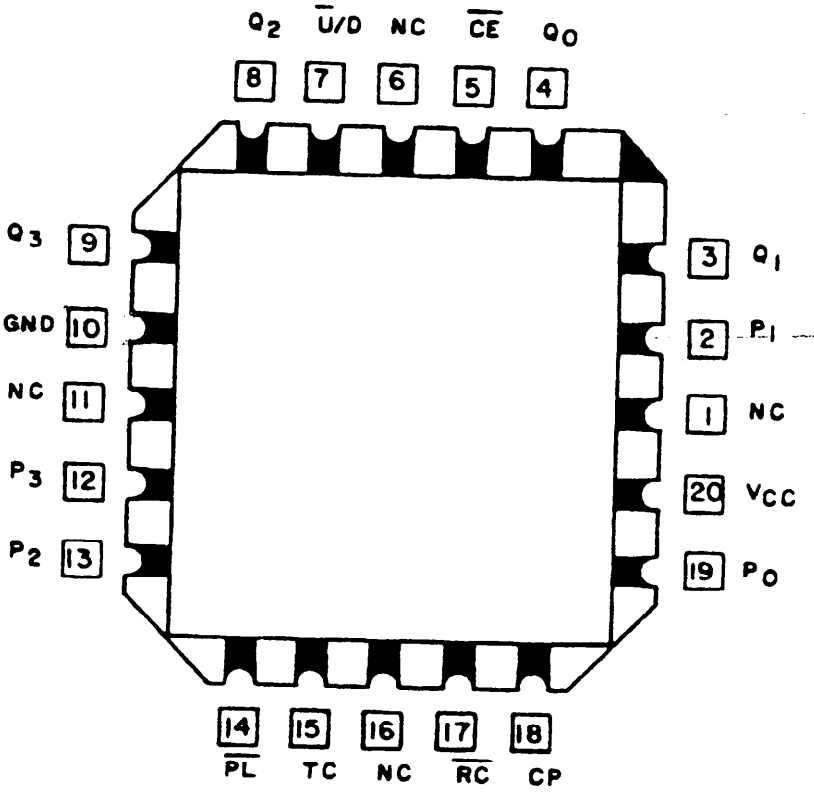


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 03

Case 2



Device type 04

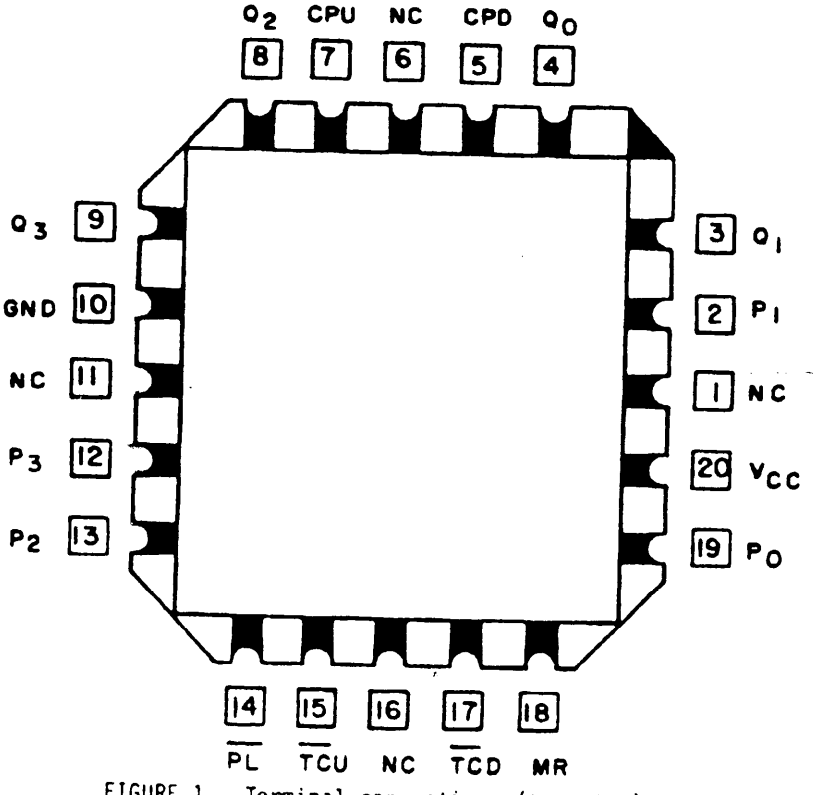


FIGURE 1. Terminal connections (top view) - Continued.

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Device types 01 and 02

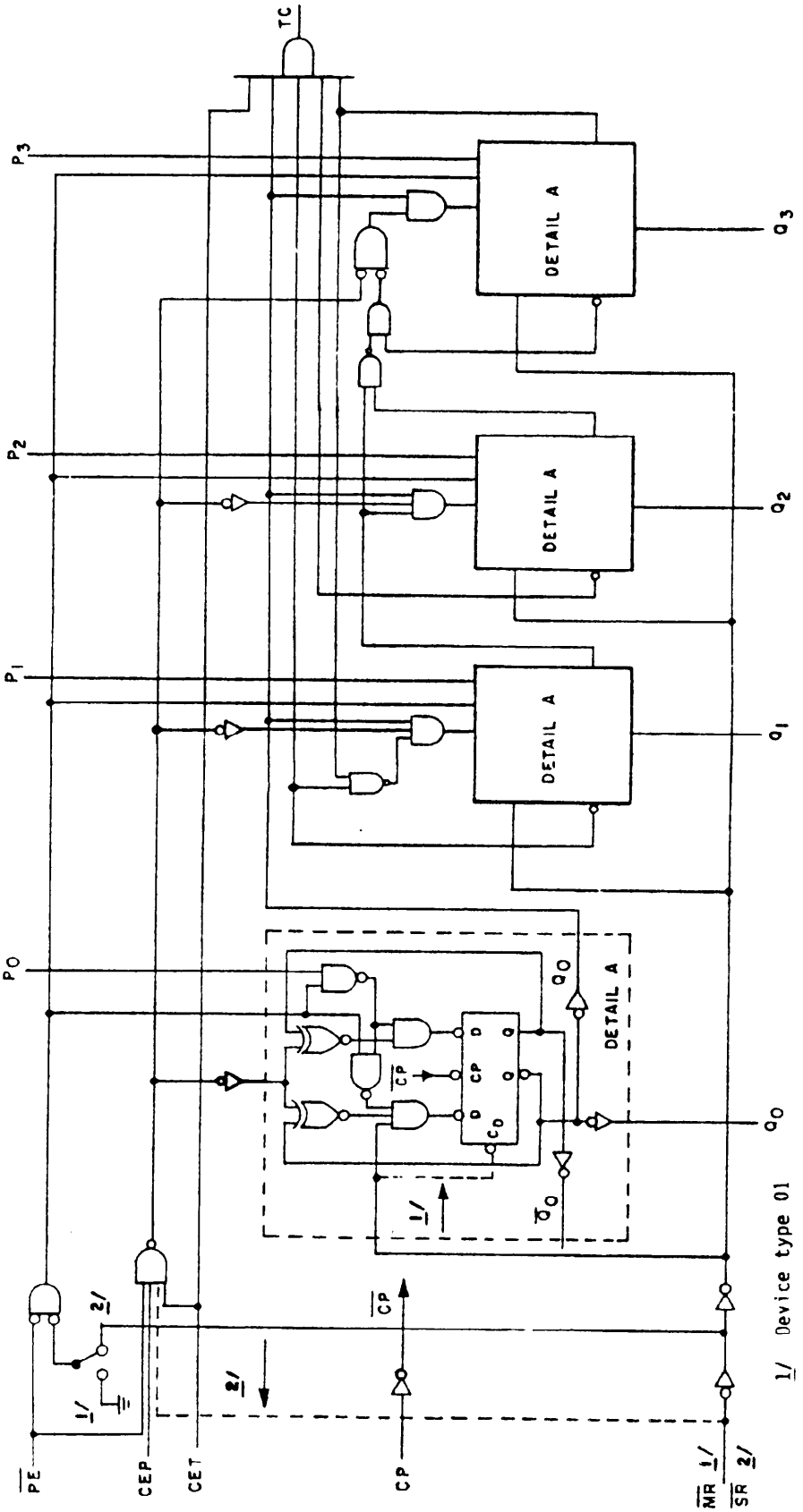


FIGURE 2. Logic diagram.

1/ Device type 01

2/ Device type 02

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Device type 03

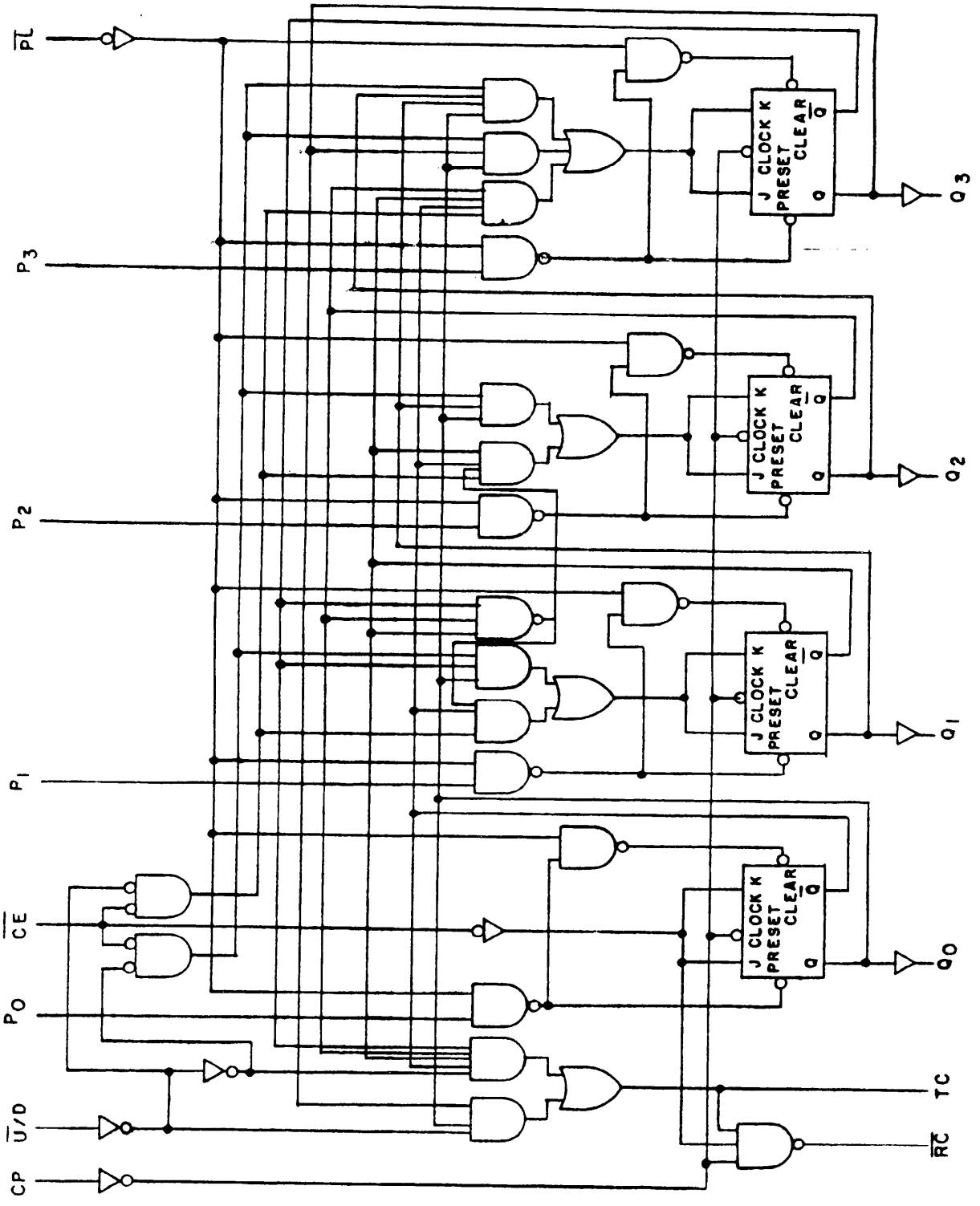


FIGURE 2. Logic diagram - Continued.

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Device type 04

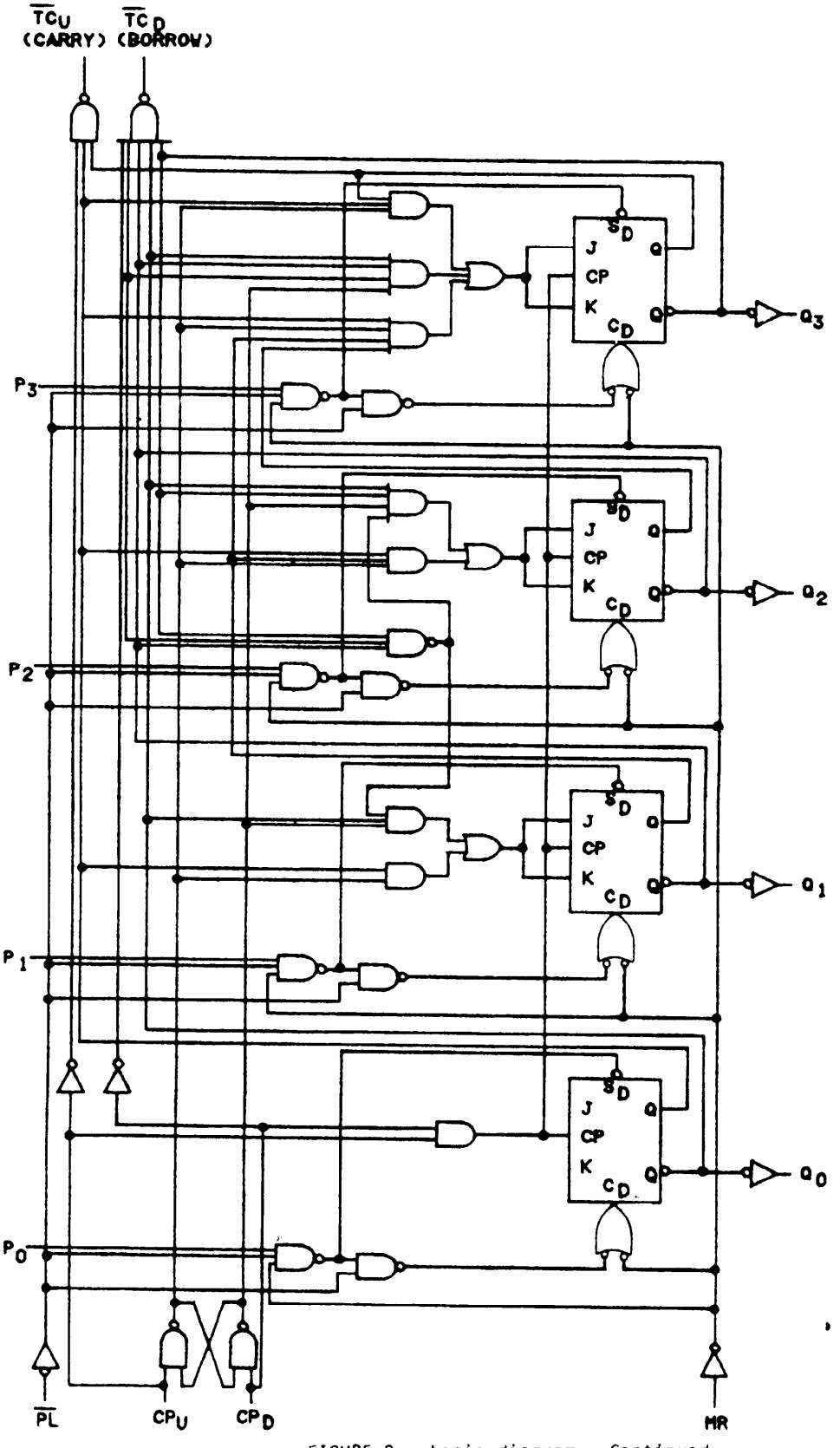


FIGURE 2. Logic diagram - Continued.

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Device types 01 and 02

Mode Select Table

*$\overline{SR}$	$\overline{PE}$	CET	CEP	Action on the Rising Clock Edge (⌐)
L	X	X	X	Reset (Clear)
H	L	X	X	Load (P_n - Q_n)
H	H	H	H	Count (Increment)
H	H	L	X	No Change (Hold)
H	H	X	L	No Change (Hold)

*For device type 02
H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

Device type 03

$\overline{RC}$ Truth Table

Inputs			Outputs
$\overline{CE}$	TC*	CP	$\overline{RC}$
L	H	⌐	⌐
H	X	X	H
X	L	X	H

*TC is generated internally
H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

Mode Select Table

Inputs				Mode
$\overline{PL}$	$\overline{CE}$	$\overline{U}$/D	CP	
H	L	L	⌐	Count Up
H	L	H	⌐	Count Down
L	X	X	X	Preset (Asyn)
H	H	X	X	No Change (Hold)

Device type 04

Function Table

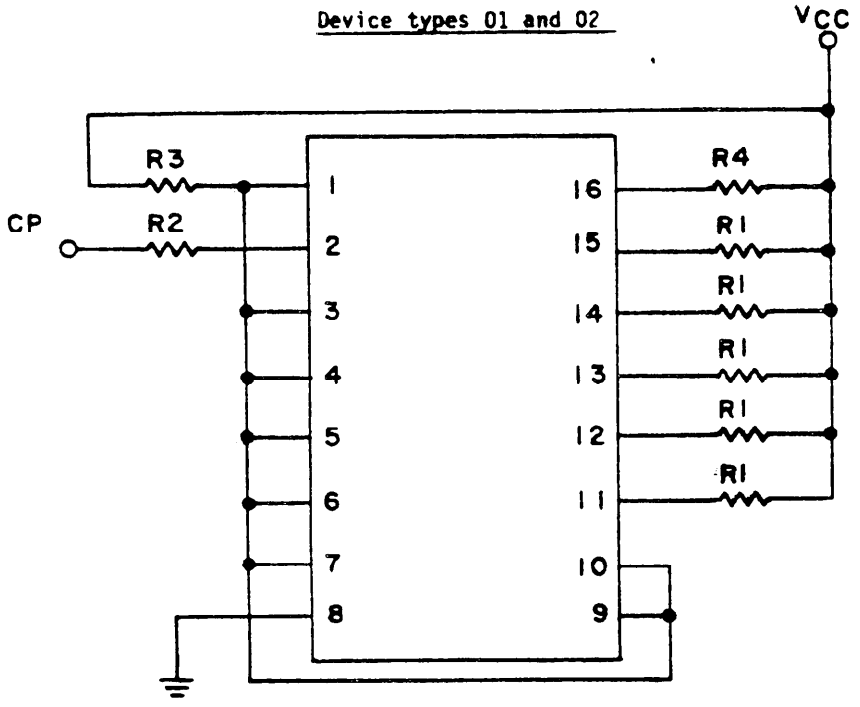
MR	$\overline{PL}$	CPU	CPD	Mode
H	X	X	X	Reset (Asyn.)
L	L	X	X	Preset (Asyn.)
L	H	H	H	No Change
L	H	⌐	H	Count Up
L	H	H	⌐	Count Down

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

FIGURE 3. Truth tables.

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Device types 01 and 02



Device type 03

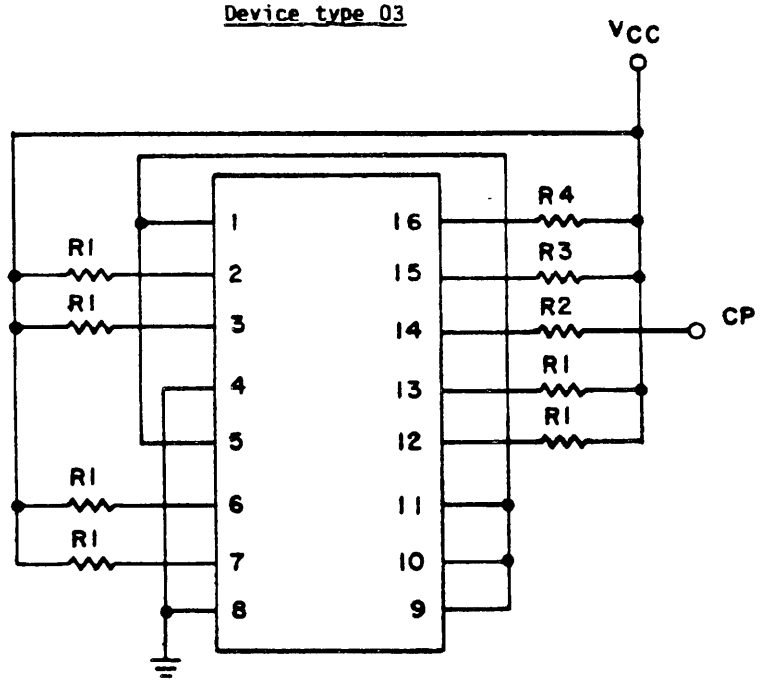
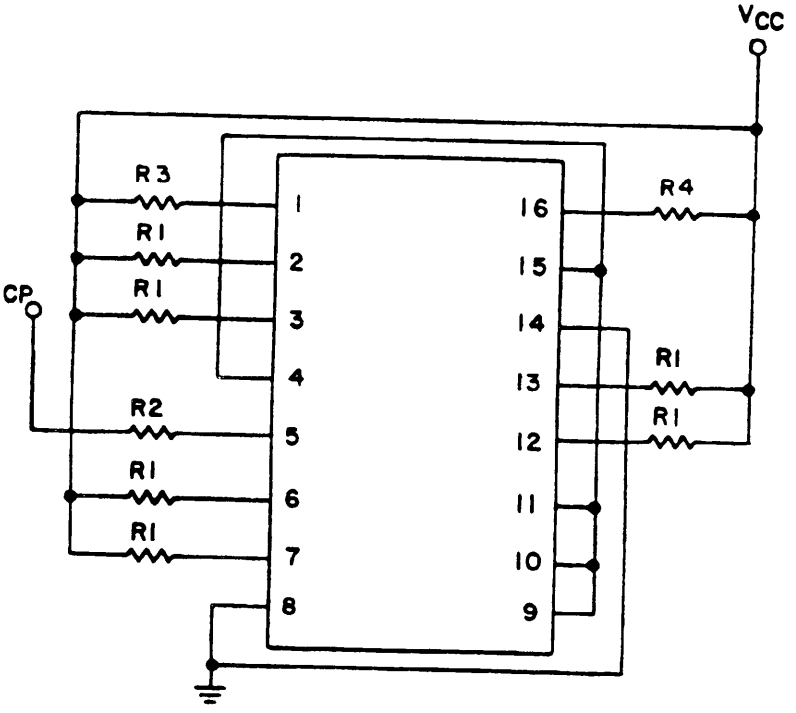


FIGURE 4. Burn-in and life test circuit.

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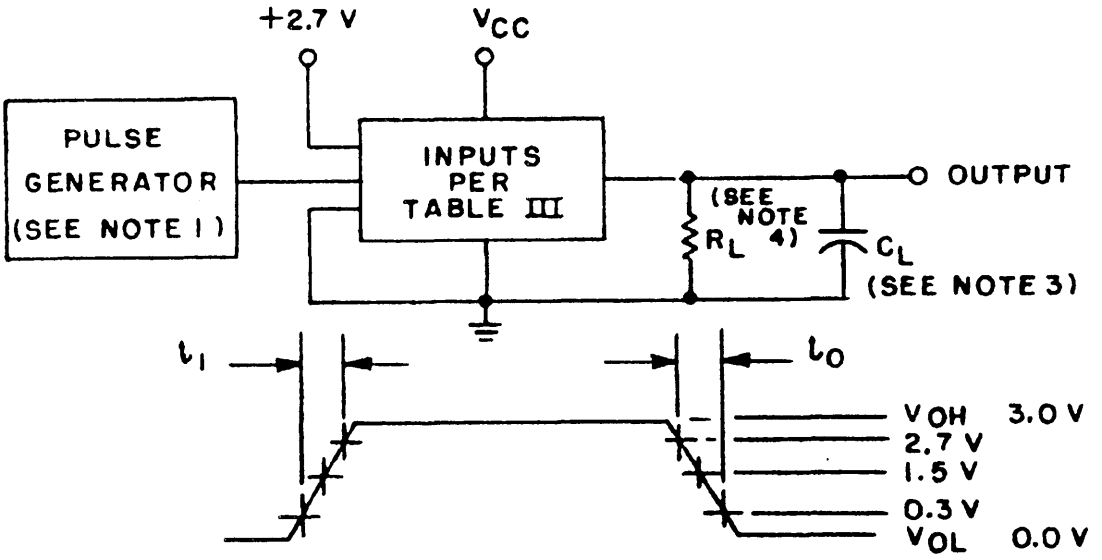
Device type 04



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

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NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. RL = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching test circuit and waveform for all device types.

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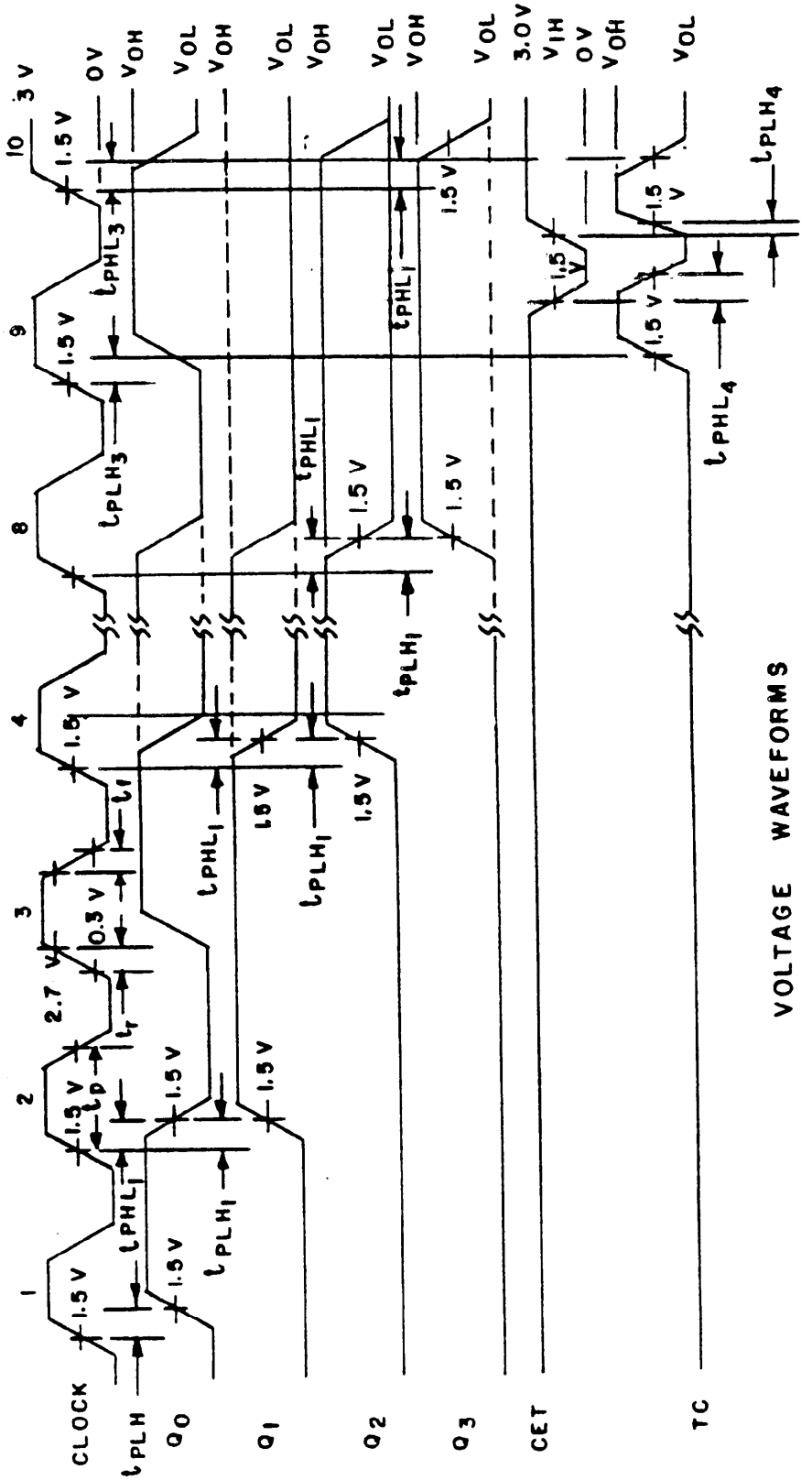
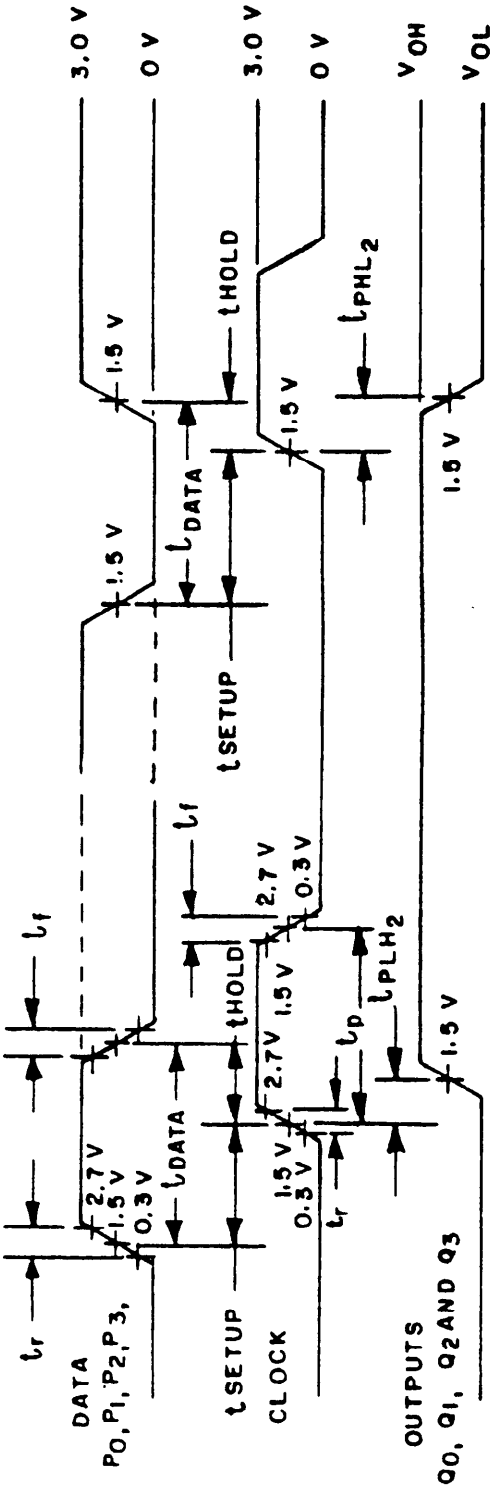


FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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NOTE: The data pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 2.5$ ns, $t_f \leq 2.5$ ns, t_{DATA} = 7.0 ns, tsetup = 5.0 ns, tHOLD = 2.0 ns.

FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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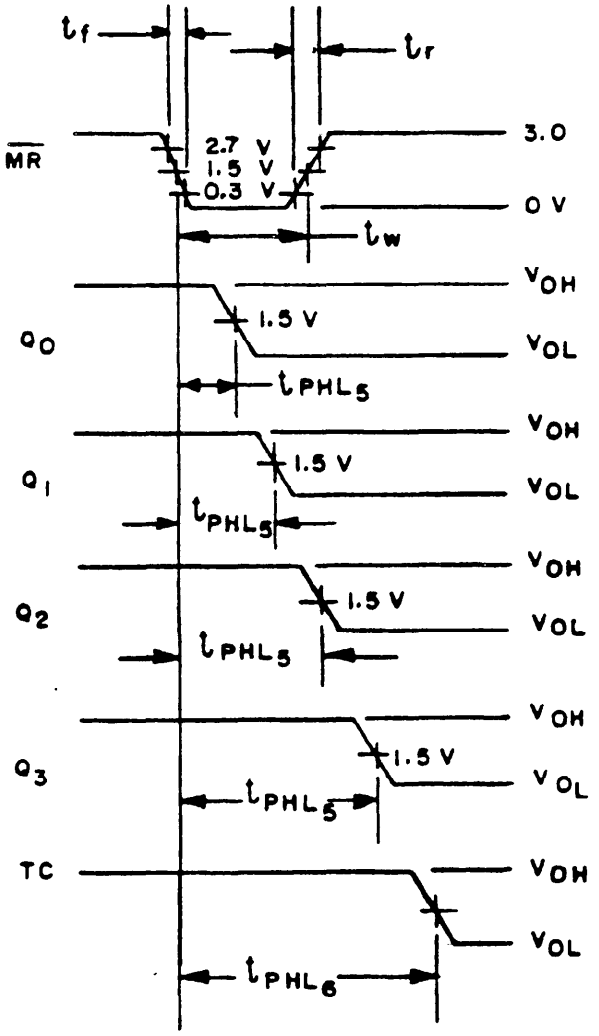
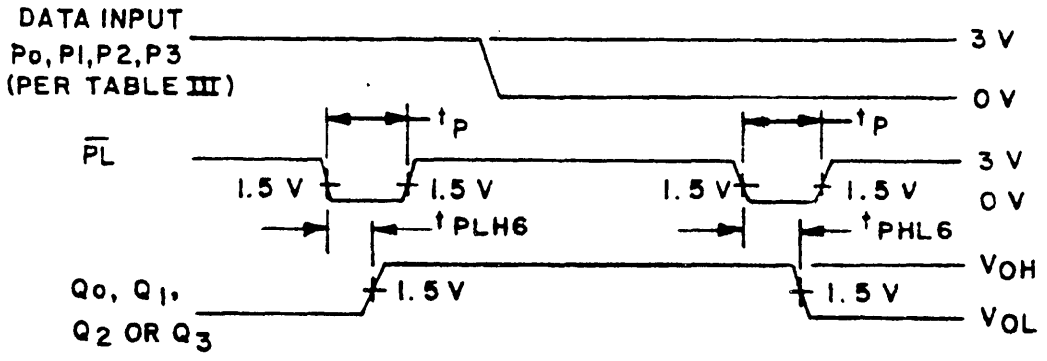
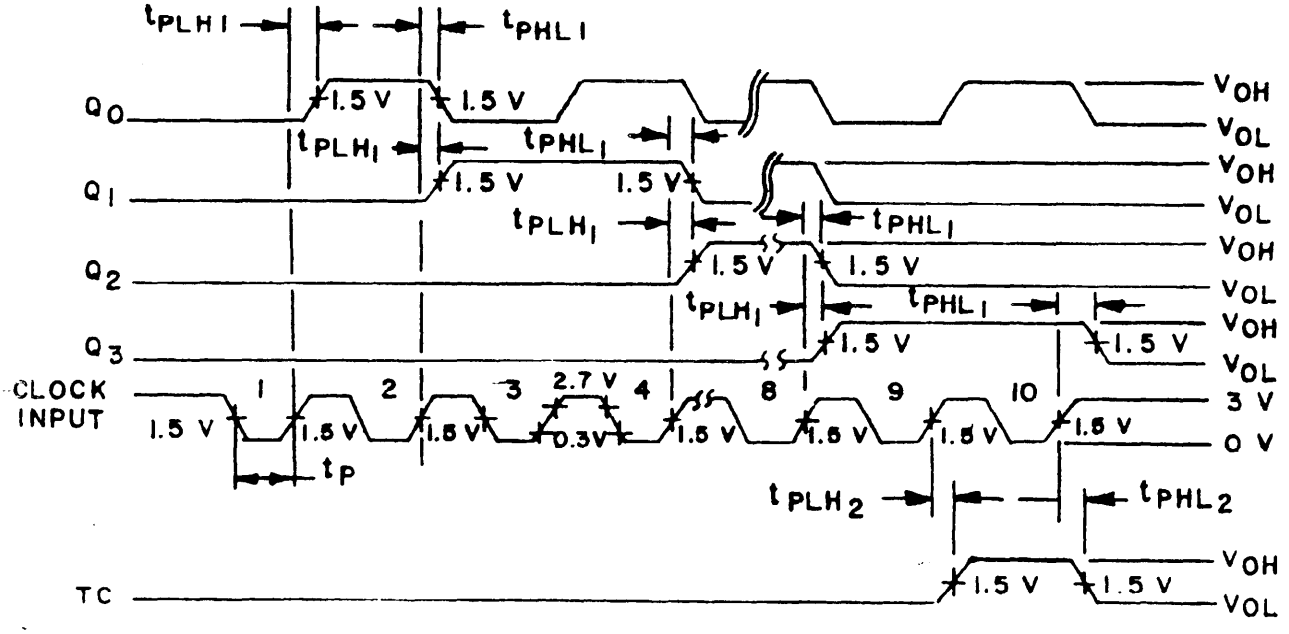


FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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PARALELL LOADED VOLTAGE WAVEFORMS



SERIAL LOADED VOLTAGE WAVEFORMS
(COUNT UP MODE)

FIGURE 5. Switching time test circuit and waveforms for device type 03 - Continued.

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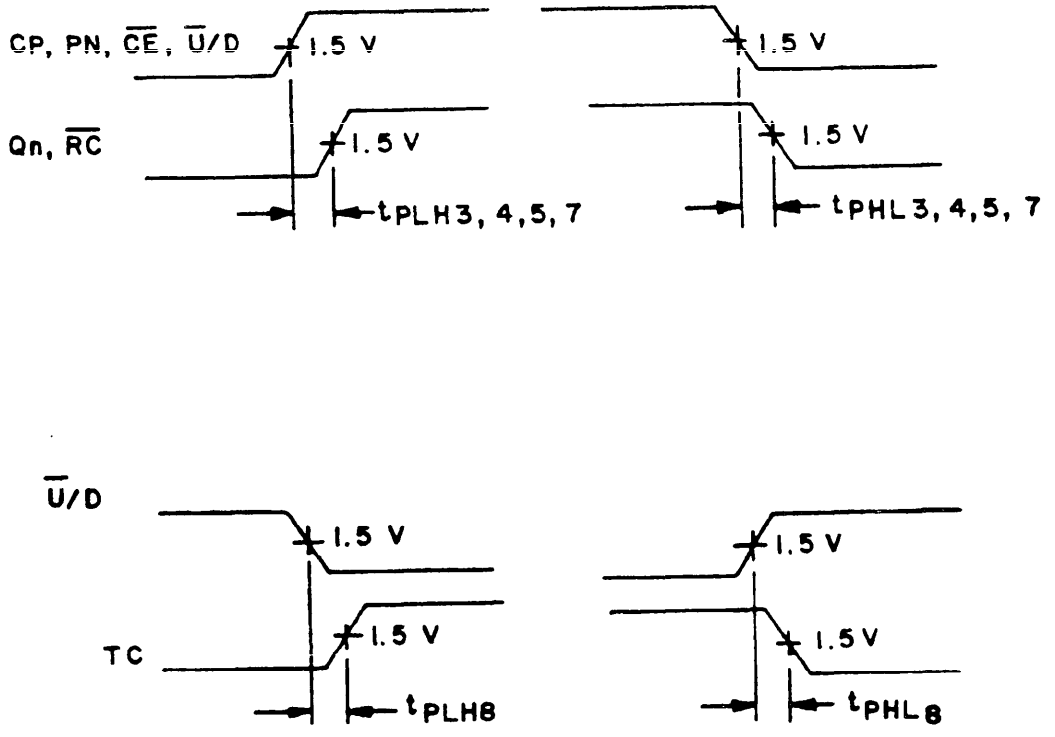
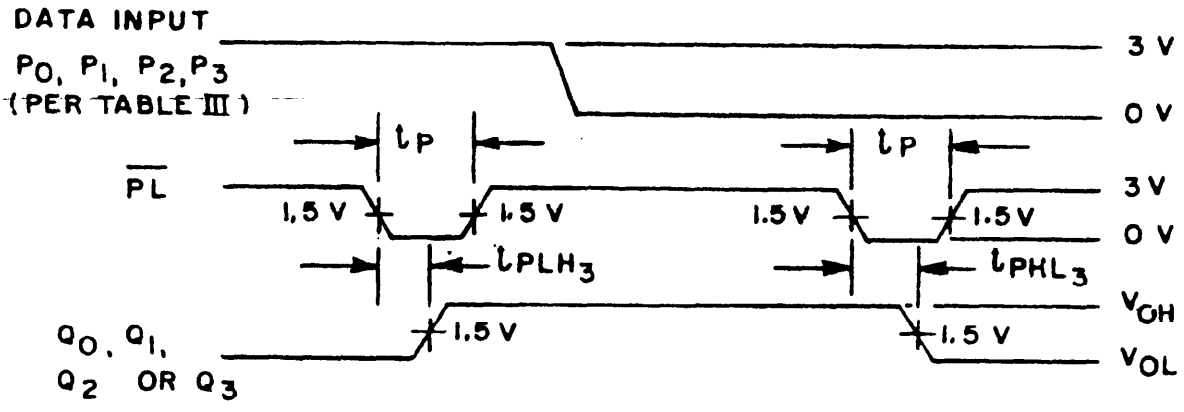
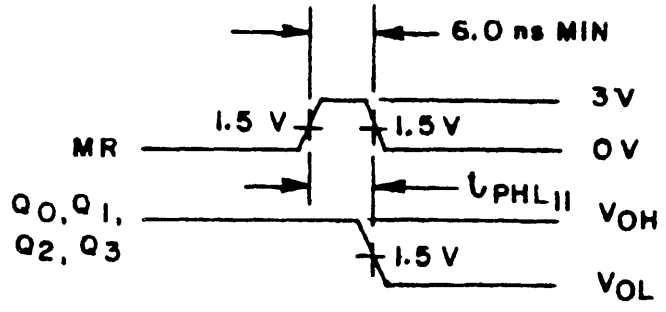


FIGURE 5. Switching time test circuit and waveforms for devive type 03 - continued.

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PARALLEL LOADED VOLTAGE WAVEFORMS



CLEAR SWITCHING VOLTAGE WAVEFORM

FIGURE 5. Switching time circuit and waveforms for device type 04 - Continued.

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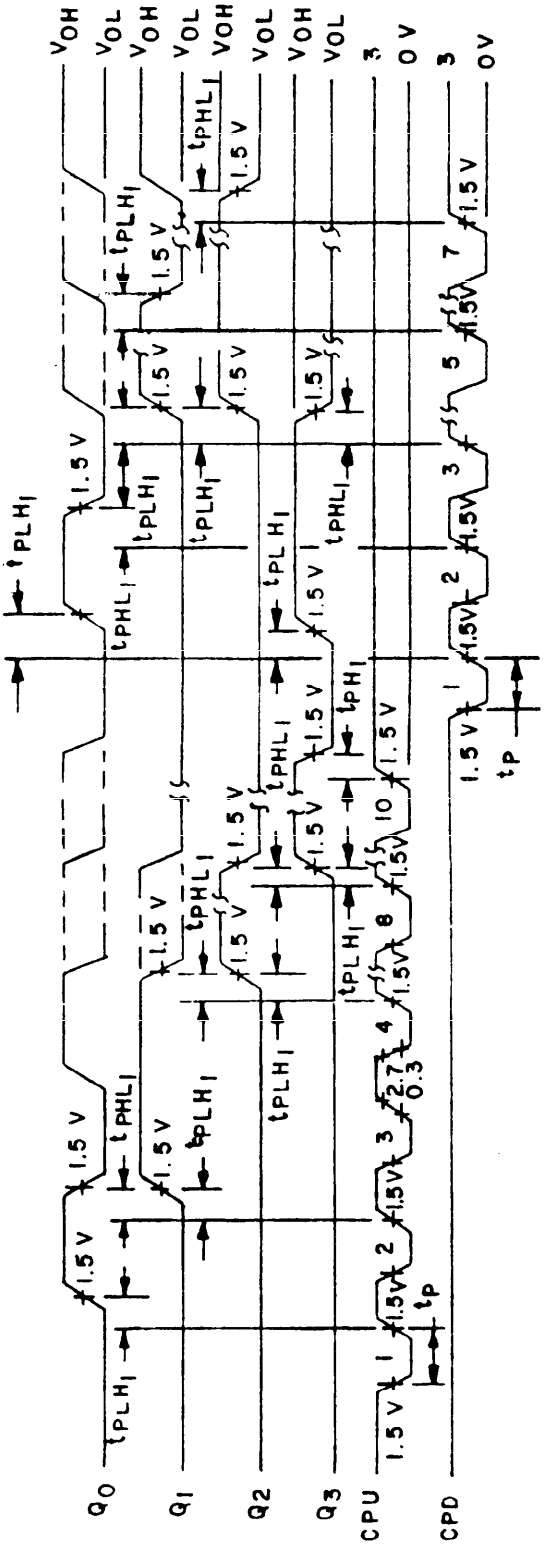


FIGURE 5. Switching time circuit and waveforms for device type 04 - Continued.

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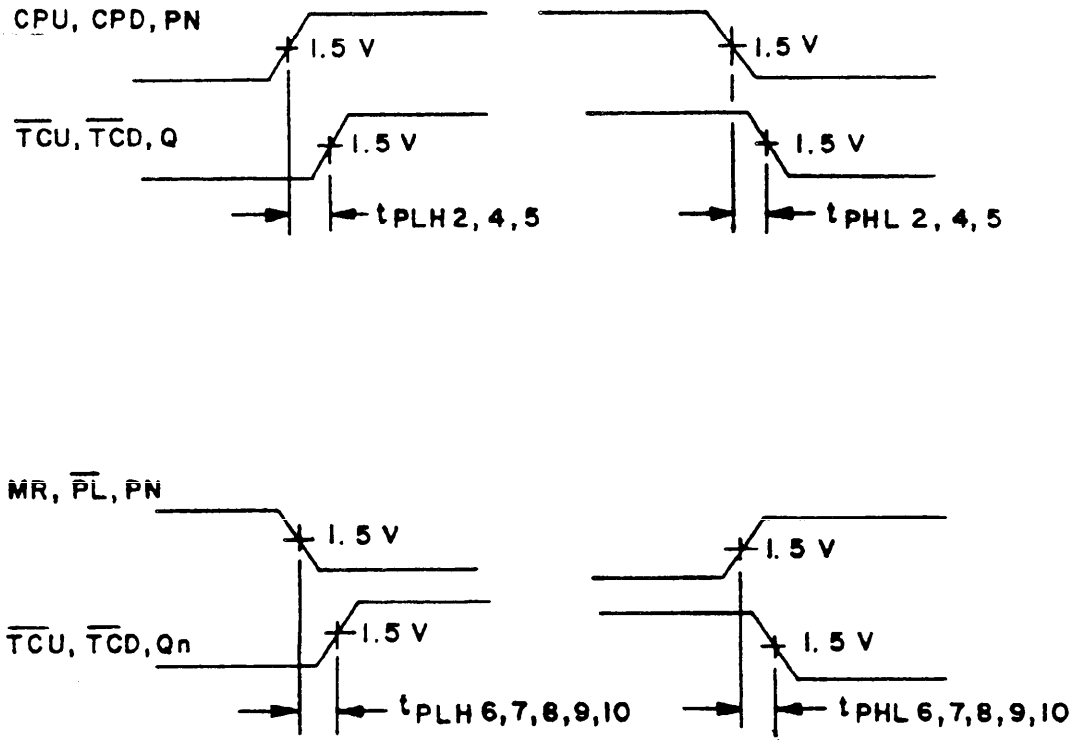


FIGURE 5. Switching time test circuit and waveforms for device type 04 - continued.

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TABLE III. Group A inspection for device 01.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 17 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits Min	Max	Unit
			Test no.	$\overline{WE}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2/	2/	2.0 V				5.5 V	GND	0.8 V	5.5 V				-1.0 mA		4.5 V	Q_0	2.5		V
			2	"	"	"	2.0 V			"	"	"	"			-1.0 mA			"	Q_1	"		"
			3	"	"	"		2.0 V		"	"	"	"		-1.0 mA				"	Q_2	"		"
			4	"	"	"			2.0 V	"	"	"	"	-1.0 mA					"	Q_3	"		"
			5	"	"	5.5 V			5.5 V	"	"	"	2.0 V					-1.0 mA	"	TC	"		"
	V_{OL}	3007	5	"	"	5.5 V			5.5 V	"	"	"	0.8 V					20 mA	"	TC		0.5	"
			7	5.5 V	"	0.8 V				"	"	"	5.5 V				20 mA		"	Q_0		"	"
			8	"	"	"	0.8 V			"	"	"	"			20 mA			"	Q_1		"	"
			9	"	"	"		0.8 V		"	"	"	"		20 mA				"	Q_2		"	"
			10	"	"	"			0.8 V	"	"	"	"	20 mA					"	Q_3		"	"
	V_{IC}		11	-18 mA	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	5.5 V	5.5 V						"	$\overline{WE}$		-1.2	V
			12	5.5 V	-18 mA	5.5 V	"	"	"	"	"	"	"						"	CP		"	"
			13	"	5.5 V	-18 mA	-18 mA	"	"	"	"	"	"						"	P_0		"	"
			14	"	"	5.5 V	"	-18 mA	"	"	"	"	"						"	P_1		"	"
			15	"	"	"	5.5 V	"	-18 mA	"	"	"	"						"	P_2		"	"
			16	"	"	"	"	5.5 V	"	-18 mA	"	"	"						"	P_3		"	"
			17	"	"	"	"	"	5.5 V	5.5 V	"	"	"						"	CEP		"	"
			18	"	"	"	"	"	"	"	"	-18 mA	"						"	$\overline{PE}$		"	"
			19	"	"	"	"	"	"	"	"	5.5 V	-18 mA						"	CET		"	"
	I_{IH1}	3010	20	2.7 V	2.7 V						"								5.5 V	$\overline{WE}$		20	µA
			21			2.7 V					"								"	CP		"	"
			22				2.7 V				"	5.5 V							"	P_0		"	"
			23					2.7 V			"	"							"	P_1		"	"
			24						2.7 V		"	"							"	P_2		"	"
			25							2.7 V	"	"							"	P_3		"	"
			26								"	5.0 V	10.0 V						"	CEP		"	"
			27								"	2.7 V	10.0 V						"	$\overline{PE}$		"	"
			28								"	5.0 V	2.7 V						"	CET		"	"
	I_{IH2}		29	7.0 V	7.0 V						"								"	$\overline{WE}$		100	µA
			30								"								"	CP		"	"
			31			7.0 V					"	5.5 V							"	P_0		"	"
			32				7.0 V				"	"							"	P_1		"	"
			33					7.0 V			"	"							"	P_2		"	"
			34						7.0 V		"	"							"	P_3		"	"
			35							7.0 V	"	5.0 V	10.0 V						"	CEP		"	"
			36							10.0 V	"	7.0 V	10.0 V						"	$\overline{PE}$		"	"
			37							10.0 V	"	5.0 V	7.0 V						"	CET		"	"
	I_{IL1}	3009	38	0.5 V	0.5 V						"								"	$\overline{WE}$	3/	3/	mA
			39			0.5 V					"								"	CP	"	"	"
			40				0.5 V				"								"	P_0	"	"	"
			41					0.5 V			"								"	P_1	"	"	"
			42						0.5 V		"								"	P_2	"	"	"
			43							0.5 V	"								"	P_3	"	"	"
			44							0.5 V	"	5.5 V	5.5 V						"	CEP	"	"	"
	I_{IL2}		45							5.5 V	"	0.5 V	5.5 V						"	$\overline{PE}$	"	"	"
			46 6/	5.5 V						5.5 V	"	5.5 V	0.5 V						"	CET	"	"	"
	I_{OS}	3011	47	2/	2/	5.5 V			5.5 V	"	"		5.5 V				10.0 V		"	Q_0	-60	-150	mA
			48	"	"		5.5 V			"	"		"			10.0 V			"	Q_1	"	"	"
			49	"	"			5.5 V		"	"		"		10.0 V				"	Q_2	"	"	"
			50	"	"				5.5 V	"	"		"	10.0 V					"	Q_3	"	"	"
			51	"	"	5.5 V			5.5 V	"	"		"					10.0 V	"	TC	"	"	"
	I_{CC}	3005	52	5.5 V	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	5.5 V	5.5 V						"	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of table.

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TABLE III. Group A inspection for device 01 - Continued.
Terminal conditions (pins not designated may be High 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				WE	CP	P_0	P_1	P_2	P_3	CEP	GND	WE	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}								
7 T_C = +25°C	Functional 4/	3014	53	B	B	B	B	B	B	B	A	A	B	L	L	L	L	L	5/					All outputs			
			54	A	A	B	B	B	B	A	A	A	A	L	L	L	H	L									
			55			A											L										
			56			B											H										
			57			A											L										
			58			B											H										
			59			A											L										
			60			B											H										
			61			A											L										
			62			B											H										
			63			A											L										
			64			B											H										
			65			A											L										
			66			B											H										
			67			A											L										
			68			B											H										
			69			A											L										
			70			B											H										
			71			A											L										
			72			B											H										
			73			A											L										
			74			B											H										
			75			A											L										
			76			B											H										
			77			A											L										
			78			B											H										
			79			A											L										
			80			B											H										
			81			A											L										
			82			B											H										
			83			A											L										
			84			B											H										
			85			A											L										
			86			B											H										
			87			A											L										
			88			B											H										
			89			A											L										
			90			B											H										
			91			A											L										
			92			B											H										
			93			A											L										
			94			B											H										
			95			A											L										
			96			B											H										
			97			A											L										
			98			B											H										
			99			A											L										
			100			B											H										
			101			A											L										
			102			B											H										
			103			A											L										
			104			B											H										
			105			A											L										
			106			B											H										
			107			A											L										
			108			B											H										
			109			A											L										
			110			B											H										
			111			A											L										
			112			B											H										
			113			A											L										
			114			B											H										
			115			A											L										
			116			B											H										
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																										

See footnotes at end of table.

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TABLE III. Group A inspection for device 01 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				$\overline{PE}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
9 T_C = +25°C	f_{MAX} 3/	3003 Fig. 5	117	2.7 V	IN					2.7 V	GND	2.7 V	2.7 V			MIT	OUT		5.0 V	Q_0	90		MHz
			118	"	"					"	"	"	"						"	Q_1	"		"
			119	"	"					"	"	"	"						"	Q_2	"		"
			120	"	"					"	"	"	"	OUT	OUT				"	Q_3	"		"
	t_{PLH1}		121 7/	2/	-					-	-	-	-						-	CP to Q_0	13.5	17.5	ns
			122 "	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			123 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			124 "	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_3	"	"	"
	t_{PHL1}		125 "	2.7 V	"					"	"	"	"						-	CP to Q_0	"	10.0	"
			126 "	2/	"					"	"	"	"						"	CP to Q_1	"	"	"
			127 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			128 "	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_3	"	"	"
	t_{PLH2}		129	"	"	2.7 V				"	"		"						-	CP to Q_0	"	8.5	"
			130	2.7 V	"		2.7 V			"	"		"						"	CP to Q_1	"	"	"
			131	"	"			2.7 V		"	"		"						"	CP to Q_2	"	"	"
			132	"	"				2.7 V	"	"		"	OUT	OUT	OUT	OUT		"	CP to Q_3	"	"	"
	t_{PHL2}		133	"	"					"	"		"						-	CP to Q_0	"	"	"
			134	"	"					"	"		"						"	CP to Q_1	"	"	"
			135	"	"					"	"		"						"	CP to Q_2	"	"	"
			136	"	"					"	"		"	OUT	OUT	OUT	OUT		"	CP to Q_3	"	"	"
	t_{PLH3}		137	2/	"				2.7 V	"	"	2/	"					OUT	"	CP to TC	14.5	18.0	"
	t_{PHL3}		139	2.7 V	"					"	"	2.7 V	"					OUT	"	CP to TC	14.5	14.0	"
	t_{PLH4}		139	"	2/	2.7 V			2.7 V	"	"	0.0 V	IN					OUT	"	CET to TC	12.5	17.5	"
	t_{PHL4}		140	"	"	"				"	"	0.0 V	IN					OUT	"	CET to TC	12.5	17.5	"
	t_{PHL5}		141	IN	"	-				-	-		2.7 V						-	MR to Q_0	15.5	12.5	ns
			142	"	"		2.7 V			"	"		"						"	MR to Q_1	"	"	"
			143	"	"			2.7 V		"	"		"						"	MR to Q_2	"	"	"
			144	"	"				2.7 V	"	"		"	OUT	OUT	OUT	OUT		"	MR to Q_3	"	"	"
	t_{PHL6}		145	"	"	2.7 V			2.7 V	"	"		"					OUT	"	MR to TC	14.5	10.5	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits for table I.																						
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of table.

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TABLE III. Group A inspection for device 02.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				SR	CP	P_0	P_1	P_2	P_3	CEP	GND	PE	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	5.5 V	2/	12.0 V	2.0 V				GND	0.8 V					-1.0 mA		4.5 V	Q_0	2.5		V
			2	"	"	15.5 V	5.5 V	2.0 V			"	"		-1.0 mA		-1.0 mA	"		"	Q_1	"		"
			3	"	"	"	5.5 V	5.5 V	2.0 V		"	"			-1.0 mA		"		"	Q_2	"		"
			4	"	"	"			5.5 V		"	"				-1.0 mA	"		"	Q_3	"		"
			5	"	"	"					"	"	2.0 V					-1.0 mA	"	TC	"		"
	V_{OL}	3007	6	"	"	"			0.8 V		"	0.8 V		20 mA					"	Q_3		0.5	"
			7	"	"	"		0.8 V	5.5 V		"	"			20 mA				"	Q_2		"	"
			8	"	"	"	0.8 V	5.5 V	"		"	"				20 mA			"	Q_1		"	"
			9	"	"	"	5.5 V	"	"	5.5 V	"	"	0.9 V				20 mA		"	Q_0		"	"
			10	"	"	"		5.5 V	"		"	"						20 mA	"	TC		"	"
	V_{IC}		11	-18 mA	5.5 V	5.5 V	5.5 V					5.5 V	5.5 V						5.5 V	SR		-1.2	V
			12	5.5 V	-18 mA	"	"	5.5 V				"	"						"	CP		"	"
			13	"	5.5 V	-18 mA	5.5 V	"	5.5 V			"	"						"	P_0		"	"
			14	"	"	5.5 V	-18 mA	5.5 V	"			"	"						"	P_1		"	"
			15	"	"	"	5.5 V	-18 mA	5.5 V			"	"						"	P_2		"	"
			16	"	"	"	"	5.5 V	-18 mA			"	"						"	P_3		"	"
			17	"	"	"	"	"	5.5 V	-18 mA		"	"						"	CEP		"	"
			18	"	"	"	"	"	"			-18 mA	-18 mA						"	PE		"	"
			19	"	"	"	"	"	"										"	CET		"	"
	I_{IH1}	3313	20	2.7 V	2.7 V															SR		20	uA
			21			2.7 V														CP		"	"
			22				2.7 V													P_0		"	"
			23					2.7 V												P_1		"	"
			24						2.7 V											P_2		"	"
			25							2.7 V										P_3		"	"
			26									5.5 V								CEP		"	"
			27									0.0 V	0.0 V							PE		"	"
			28									2.7 V	0.0 V							CET		"	"
			29									0.0 V	2.7 V										
	I_{IH2}		30	7.0 V	7.0 V															SR		100	uA
			31			7.0 V														CP		"	"
			32				7.0 V													P_0		"	"
			33					7.0 V												P_1		"	"
			34						7.0 V											P_2		"	"
			35							7.0 V										P_3		"	"
			36									5.5 V								CEP		"	"
			37									0.0 V	0.0 V							PE		"	"
												0.0 V	7.0 V							CET		"	"
	I_{IL1}	3309	38			0.5 V														CP	3/	3/	mA
			39				0.5 V													P_0	"	"	"
			40					0.5 V												P_1	"	"	"
			41						0.5 V											P_2	"	"	"
			42							0.5 V										P_3	"	"	"
			43	5.5 V																CEP	"	"	"
												5.5 V	5.5 V										
	I_{IL2}		44	0.5 V																SR			
			45	5.5 V																PE			
			46 6/	5.5 V																CET			
										5.5 V		5.5 V	5.5 V										
												0.5 V	5.5 V										
												5.5 V	0.5 V										
	I_{OS}	3011	47		2/									0.0 V						Q_3	-60	-150	
			48						5.5 V						0.0 V					Q_2			
			49					5.5 V								0.0 V				Q_1			
			50				5.5 V										0.0 V			Q_0			
			51				5.5 V			5.5 V			5.5 V					0.0 V		TC			
	I_{CC}	3005	52	5.5 V	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	5.5 V	"	5.5 V	5.5 V						"	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of table.

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TABLE III. Group A inspection for device 02 - Continued.
Terminal conditions (pins not designated may be High ≥ 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				SR	CP	P_0	P_1	P_2	P_3	CEP	GND	PE	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
7 T_C = +25°C	Func-tional 4/	3014	53	B	2/	B	B	B	B	B	GND	A	B	L	L	L	L	L	3/	All outputs			
			54	A						A			A										
			55		B	A																	
			56																				
			57		A	B																	
			58																				
			59		B	A																	
			60																				
			61		A	B																	
			62																				
			63		B	A																	
			64																				
			65		A	B																	
			66																				
			67		B	A																	
			68																				
			69		A	B																	
			70																				
			71		B	A																	
			72																				
			73		A	B																	
			74																				
			75		B	A																	
			76																				
			77		A	B																	
			78																				
			79		B	A																	
			80																				
			81		A	B																	
			82																				
			83		B	A																	
			84																				
			85		A	B																	
			86																				
			87		B	A																	
			88																				
			89		A	B																	
			90																				
			91		B	A																	
			92																				
			93		A	B																	
			94																				
			95		B	A																	
			96																				
			97		A	B																	
			98																				
			99		B	A																	
			100																				
			101		A	B																	
			102																				
			103		B	A																	
			104																				
			105		A	B																	
			106																				
			107		B	A																	
			108																				
			109		A	B																	
			110																				
			111		B	A																	
			112																				
			113		A	B																	
			114																				
			115		B	A																	
			116																				
			117		A	B																	
			118		B	B																	
			119																				

See footnotes at end of table.

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TABLE III. Group A inspection for device 02 - Continued.
Terminal conditions (pins not designated may be High 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{SR}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
			120	A	A	3	A	A	H	A	GND	9	A	L	L	L	L	L	9/	All outputs			
			121	"	3	"	"	"	"	"	"	"	"	L	L	L	"	"	"	"			
			122	"	A	"	"	"	"	"	"	"	"	H	H	H	"	"	"	"			
			123	"	A	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"			
			124	"	9	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			125	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			126	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			127	"	A	"	"	"	"	"	"	"	9	L	L	L	H	L	"	"			
9	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 3/	3003 Fig. 5	128	2.7 V	IN					2.7 V	GND	2.7 V	2.7 V	OUT	OUT	OUT	OUT		5.0 V	Q_0	90		MHz
			129	"	"					"	"	"	"						"	Q_1	"		"
			130	"	"					"	"	"	"						"	Q_2	"		"
			131	"	"					"	"	"	"						"	Q_3	"		"
	t_{PLH1}		132 7/	2/	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	3.5	7.5	ns
			133 "	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			134 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			135 "	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PHL1}		136 "	2.7 V	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	"	10.5	"
			137 "	2/	"					"	"	"	"						"	CP to Q_1	"	"	"
			138 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			139 "	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PLH2}		140	2.7 V	"	2.7 V				"	"	0.0 V	0.0 V	OUT	OUT	OUT	OUT		"	CP to Q_0	"	18.5	"
			141	"	"		2.7 V			"	"	"	"						"	CP to Q_1	"	"	"
			142	"	"			2.7 V		"	"	"	"						"	CP to Q_2	"	"	"
			143	"	"				2.7 V	"	"	"	"						"	CP to Q_3	"	"	"
	t_{PHL2}		144	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	"	"	"
			145	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			146	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			147	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PLH3}		148	"	"	0.0 V	0.0 V	0.0 V	2.7 V	"	"	2/	2.7 V					OUT	"	CP to TC	14.5	16.0	"
	t_{PHL3}		149	"	"					"	"	2.7 V	2.7 V					"	"	CP to TC	14.5	16.0	"
	t_{PLH4}		150	"	2/	2.7 V			2.7 V	"	"	0.0 V	IN					"	"	CET to TC	12.5	17.5	"
	t_{PHL4}		151	"	"	"			2.7 V	"	"	0.0 V	IN					"	"	CET to TC	12.5	17.5	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits for table 1.																						
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of table.

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TABLE III. Group A inspection for device 01.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 P_1	2 3 $\overline{Q}_1$	3 4 Q_0	4 5 $\overline{CE}$	5 7 $D/\overline{D}$	6 8 Q_2	7 9 Q_3	8 10 GND	9 12 P_3	10 13 P_2	11 14 $\overline{PL}$	12 15 TC	13 17 $\overline{RC}$	14 19 CP	15 19 P_0	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	-1.0 mA		2.0 V				GND			0.8 V					4.5 V	Q_1	2.5		V
			2			-1.0 mA	"				"			"				2.0 V	"	$\overline{Q}_0$	"		"
			3				"		-1.0 mA		"		2.0 V	"					"	Q_2	"		"
			4				"			-1.0 mA	"	2.0 V	"	"					"	Q_3	"		"
			5	0.8 V			0.8 V	0.8 V			"	"	0.8 V	"	-1.0 mA		2.0 V	2.0 V	"	TC	"		"
			6	0.8 V			0.8 V	0.8 V			"	"	0.8 V	"		-1.0 mA	2.0 V	2.0 V	"	$\overline{RC}$	"		"
	V_{OL}	3007	7	0.8 V	20 mA		2.0 V				"			"					"	Q_1		0.5	"
			8			20 mA	"				"			"				0.8 V	"	$\overline{Q}_0$		"	"
			9				"		20 mA		"		0.8 V	"					"	Q_2		"	"
			10				"			20 mA	"	0.8 V	"	"					"	Q_3		"	"
			11	0.8 V			"	2.0 V			"	2.0 V	0.8 V	"	20 mA		0.8 V	2.0 V	"	TC		"	"
			12	0.8 V			0.8 V	0.8 V			"	2.0 V	0.8 V	"		20 mA	0.8 V	2.0 V	"	$\overline{RC}$		"	"
	V_{IC}		13	-18 mA							"								"	P_1		-1.2	"
			14				-18 mA				"								"	$\overline{CE}$		"	"
			15					-18 mA			"								"	$D/\overline{D}$		"	"
			16								"	-18 mA							"	P_3		"	"
			17								"		-18 mA						"	P_2		"	"
			18								"			-18 mA					"	$\overline{PL}$		"	"
			19								"						-18 mA		"	$\overline{RC}$		"	"
			20								"							-18 mA	"	CP		"	"
	I_{IH1}	3010	21	2.7 V							"			5.5 V					5.5 V	P_1		20	µA
			22				2.7 V				"			"					"	$\overline{CE}$		"	"
			23					2.7 V			"			"					"	$D/\overline{D}$		"	"
			24								"	2.7 V		5.5 V					"	P_3		"	"
			25								"		2.7 V	5.5 V					"	P_2		"	"
			26								"			2.7 V					"	$\overline{PL}$		"	"
			27								"						2.7 V		"	$\overline{RC}$		"	"
			28								"			5.5 V				2.7 V	"	CP		"	"
	I_{IH2}		29	7.0 V							"			5.5 V					"	P_1		100	"
			30				7.0 V				"			"					"	$\overline{CE}$		"	"
			31					7.0 V			"			"					"	$D/\overline{D}$		"	"
			32								"	7.0 V		5.5 V					"	P_3		"	"
			33								"		7.0 V	5.5 V					"	P_2		"	"
			34								"			7.0 V					"	$\overline{PL}$		"	"
			35								"						7.0 V		"	$\overline{RC}$		"	"
			36								"			5.5 V				7.0 V	"	CP		"	"
	I_{IL1}	3009	37	0.5 V							"			0.0 V					"	P_1	3/	3/	"
			38					0.5 V			"			"					"	$D/\overline{D}$		"	"
			39								"	0.5 V		0.0 V					"	P_3		"	"
			40								"		0.5 V	0.0 V					"	P_2		"	"
			41								"			0.5 V					"	$\overline{PL}$		"	"
			42								"						0.5 V		"	CP		"	"
			43								"			0.0 V				0.5 V	"	P_0		"	"
	I_{IL2}		44				0.5 V				"								"	$\overline{CE}$		"	mA
	I_{OS}	3011	45	5.5 V	0.0 V		5.5 V				"								"	Q_1	-60	-150	"
			46			0.0 V	"				"							5.5 V	"	$\overline{Q}_0$	"	"	"
			47				"		0.0 V		"		5.5 V						"	Q_2	"	"	"
			48				"			0.0 V	"	5.5 V							"	Q_3	"	"	"
			49								"				0.0 V		5.5 V	5.5 V	"	TC	"	"	"
			50								"					0.0 V	5.5 V	5.5 V	"	$\overline{RC}$	"	"	"
	I_{CC}	3005	51	0.0 V			0.0 V	0.0 V			"	0.0 V	0.0 V	0.0 V			0.0 V	0.0 V	"	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of table.

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TABLE III. Group A inspection for device 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 P_1	2 3 Q_1	3 4 D_0	4 5 $\overline{CE}$	5 7 $\overline{D}/D$	6 8 Q_2	7 9 Q_3	8 10 GND	9 12 P_3	10 13 P_2	11 14 $\overline{PE}$	12 15 TC	13 17 $\overline{RC}$	14 18 CP	15 19 P_0	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Functional 4/	3014	52	B	L	L	B	B	L	L	GND	B	B	B	L	H	↑	B	5/	All outputs			
			53	A	H	H	↓	↓	H	H	↓	A	A	A	H	↓	↓	A	↓	↓			
			54	A	H	H	↓	↓	H	H	↓	A	A	A	H	↓	↓	A	↓	↓			
			55	3	L	L	↓	↓	L	L	↓	B	B	9	L	↓	↓	B	↓	↓			
			56	9	L	L	↓	↓	L	L	↓	9	9	4	↓	↓	↓	B	↓	↓			
			57	A	L	H	↓	↓	L	H	↓	8	8	↓	↓	↓	2/	A	↓	↓			
			58	↓	H	L	↓	↓	L	H	↓	A	A	↓	↓	↓	↓	↓	↓	↓			
			59	↓	H	A	↓	↓	L	H	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
			60	↓	L	L	↓	↓	H	L	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
			61	↓	L	H	↓	↓	H	L	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
			62	↓	H	L	↓	↓	H	L	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
			63	↓	H	H	↓	↓	H	L	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			
			64	↓	L	L	↓	↓	L	H	↓	↓	↓	↓	H	↓	↓	↓	↓	↓			
			65	↓	L	H	↓	↓	L	H	↓	↓	↓	↓	H	↓	↓	↓	↓	↓			
			66	↓	L	H	↓	↓	L	H	↓	↓	↓	↓	L	H	↓	9	↓	↓			
			67	↓	L	H	↓	↓	L	H	↓	↓	↓	↓	L	↓	↓	A	↓	↓			
			68	↓	L	H	↓	↓	L	H	↓	↓	↓	↓	L	↓	↓	2/	↓	↓			
			69	↓	L	H	A	↓	L	H	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			70	↓	H	L	3	↓	L	H	↓	↓	↓	↓	L	↓	↓	B	↓	↓			
			71	↓	H	L	↓	A	L	H	↓	↓	↓	↓	H	↓	↓	B	↓	↓			
			72	↓	H	L	↓	↓	L	H	↓	↓	↓	↓	H	↓	↓	A	↓	↓			
			73	↓	H	L	↓	↓	L	H	↓	↓	↓	↓	H	↓	↓	B	↓	↓			
			74	↓	H	L	A	↓	L	H	↓	↓	↓	↓	H	↓	↓	A	↓	↓			
			75	↓	H	L	9	↓	L	H	↓	↓	↓	↓	L	H	↓	B	↓	↓			
			76	↓	H	L	↓	↓	L	H	↓	↓	↓	↓	L	H	↓	A	↓	↓			
			77	↓	H	H	↓	↓	L	H	↓	↓	↓	↓	L	L	↓	2/	↓	↓			
			78	↓	H	H	↓	↓	L	H	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			79	↓	H	H	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			80	↓	H	H	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			81	↓	H	L	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			82	↓	H	L	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			83	↓	L	H	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			84	↓	L	H	↓	↓	H	L	↓	↓	↓	↓	L	↓	↓	↓	↓	↓			
			85	B	L	L	↓	B	↓	↓	↓	B	B	B	↓	↓	A	B	↓	↓			
			86	9	L	L	↓	B	↓	↓	↓	9	9	9	↓	↓	8	9	↓	↓			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	87		OUT	OUT	GND	GND	OUT	OUT	GND			2.7 V			IH		5.0 V	Q_0	90		MHz
			88		↓	↓	↓	↓	↓	↓	↓			↓			↓		↓	Q_1	↓		↓
			89		↓	↓	↓	↓	↓	↓	↓			↓			↓		↓	Q_2	↓		↓
			90		↓	↓	↓	↓	↓	↓	↓			↓			↓		↓	Q_3	↓		↓
	t_{PLH1}		91		OUT	OUT			OUT	OUT	↓			2/			↓		↓	CP to Q_0	3.0	7.5	ns
			92		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_1	↓	↓	↓
			93		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_2	↓	↓	↓
			94		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_3	↓	↓	↓
	t_{PHL1}		95		OUT	OUT			OUT	OUT	↓			↓			↓		↓	CP to Q_0	5.0	11.0	↓
			96		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_1	↓	↓	↓
			97		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_2	↓	↓	↓
			98		↓	↓			↓	↓	↓			↓			↓		↓	CP to Q_3	↓	↓	↓
	t_{PLH2}		99								↓	2.7 V		↓	OUT		↓		↓	CP to TC	16.0	13.0	↓
	t_{PHL2}		100								↓	↓		↓	OUT		↓		↓	CP to TC	15.0	11.0	↓
	t_{PLH3}		101								↓	↓		↓		OUT	↓		↓	CP to $\overline{RC}$	13.0	7.5	↓
	t_{PHL3}		102								↓	↓		↓		OUT	↓		↓	CP to $\overline{RC}$	13.0	17.0	↓

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 2/ test no.	1 P_1	2 Q_1	3 Q_0	4 CPD	5 CPU	6 Q_2	7 Q_3	8 GND	9 P_3	10 P_2	11 PE	12 TCU	13 TCD	14 MR	15 P_0	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
7 T_C = +25°C	Functional 4/	3014	52	A	L	L	A	A	L	L	GND	A	A	A	H	H	A	A	B/	All outputs			
			53		H				H	H				B			B						
			54	B	L				L	L		B		A				B					
			55			H								B									
			56		L	L			L	L				A									
			57		H				H	H				B									
			58		L	H			L	L			B	A	L								
			59		H				H	H				B	H								
			60		L	L			L	L				A	L								
			61		H				H	H				B	H								
			62		L	H			L	L			A	A	L								
			63		H				H	H				B	H								
			64		L	L			L	L				A	L								
			65		H				H	H				B	H								
			66		L	H			L	L			B	A	L								
			67		H				H	H				B	H								
			68		L	L			L	L				A	L								
			69		H				H	H				B	H								
			70		L	H			L	L			A	A	L								
			71		H				H	H				B	H								
			72		L	L			L	L				A	L								
			73		H				H	H				B	H								
			74		L	H			L	L			B	A	L								
			75		H				H	H				B	H								
			76		L	L			L	L				A	L								
			77		H				H	H				B	H								
			78		L	H			L	L			A	A	L								
			79	A	H				H	H				B	H								
			80		L	L			L	L				A	L								
			81		H				H	H				B	H								
			82		L	H			L	L			B	A	L								
			83		H				H	H				B	H								
			84		L	L			L	L				A	L								
			85		H				H	H				B	H								
			86		L	H			L	L			A	A	L								
			87		H				H	H				B	H								
			88		L	L			L	L				A	L								
			89		H				H	H				B	H								
			90		L	H			L	L			B	A	L								
			91	A	H				H	H				B	H								
			92		L	L			L	L				A	L								
			93		H				H	H				B	H								
			94		L	H			L	L			A	A	L								
			95		H				H	H				B	H								
			96		L	L			L	L				A	L								
			97		H				H	H				B	H								
			98		L	H			L	L			B	A	L								
			99		H				H	H				B	H								
			100		L	L			L	L				A	L								
			101		H				H	H				B	H								
			102		L	H			L	L			A	A	L								
			103		H				H	H				B	H								
			104		L	L			L	L				A	L								
			105		H				H	H				B	H								
			106		L	H			L	L			B	A	L								
			107		H				H	H				B	H								
			108		L	L			L	L				A	L								
			109		H				H	H				B	H								
			110		L	H			L	L			A	A	L								
			111		H				H	H				B	H								
			112		L	L			L	L				A	L								
			113		H				H	H				B	H								
			114		L	H			L	L			B	A	L								
			115		H				H	H				B	H								
			116		L	L			L	L				A	L								
			117		H				H	H				B	H								
			118		L	H			L	L			B	A	L								
			119	L	H				H	H				B	H								

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TABLE III. Group A inspection for device 04.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 P_1	2 3 Q_1	3 4 Q_0	4 5 CPD	5 7 CPU	6 8 Q_2	7 9 Q_3	8 10 GND	9 12 P_3	10 13 P_2	11 14 PL	12 15 TCU	13 17 TCD	14 18 MR	15 19 P_0	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	-1.0 mA						GND			0.8 V			0.0 V		4.5 V	Q_1	2.5		V
			2			-1.0 mA			-1.0 mA									2.0 V		Q_2			
			3							-1.0 mA			2.0 V							Q_3			
			4									2.0 V					0.8 V	5.5 V		TCU			
			5	5.5 V			2.0 V	2.0 V				5.5 V	5.5 V		-1.0 mA	-1.0 mA	0.8 V			TCD			
			6																				
	V_{OL}	3007	7	0.8 V	20 mA									0.8 V			0.0 V			Q_1		0.5	V
			8			20 mA			20 mA									0.8 V		Q_2			
			9							20 mA			0.8 V							Q_3			
			10									0.8 V	5.5 V					5.5 V		TCU			
			11	5.5 V			0.8 V	0.8 V				5.5 V	0.8 V		20 mA	20 mA	0.8 V			TCD			
			12	0.8 V								0.8 V						0.8 V					
	V_{IC}		13	-18 mA			-18 mA													P_1		-1.2	V
			14					-18 mA												CPD			
			15																	CPU			
			16									-18 mA								P_3			
			17										-18 mA							P_2			
			18											-18 mA						PL			
			19														-18 mA			MR			
			20															-18 mA		P_0			
	I_{IH1}	3010	21	2.7 V															5.5 V	P_1		20	μA
			22				2.7 V													CPD			
			23					2.7 V												CPU			
			24									2.7 V								P_3			
			25										2.7 V							P_2			
			26											2.7 V						PL			
			27														2.7 V			MR			
			28															2.7 V		P_0			
	I_{IH2}		29	7.0 V																P_1		100	μA
			30				7.0 V													CPD			
			31					7.0 V												CPU			
			32									7.0 V								P_3			
			33										7.0 V							P_2			
			34											7.0 V						PL			
			35														7.0 V			MR			
			36															7.0 V		P_0			
	I_{IL1}	3009	37	0.5 V										0.0 V			0.0 V			P_1	3/	3/	mA
			38																	P_3			
			39									0.5 V								P_2			
			40										0.5 V	0.5 V			0.5 V			PL			
			41															0.5 V		MR			
			42																	P_0			
	I_{IL2}		43				0.5 V	5.5 V												CPD			
			44				5.5 V	0.5 V												CPU			
	I_{OS}	3011	45	5.5 V	0.0 V									0.0 V			0.0 V			Q_1	-60	-150	mA
			46			0.0 V			0.0 V									5.5 V		Q_2			
			47							0.0 V			5.5 V							Q_3			
			48									5.5 V			0.0 V					TCU			
			49					5.5 V				5.5 V				0.0 V		5.5 V		TCD			
			50				5.5 V																
	I_{CC}	3005	51	5.5 V			5.5 V	5.5 V			"	5.5 V	5.5 V	0.0 V			0.0 V	5.5 V	"	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 P_1	2 3 Q_1	3 4 Q_0	4 5 CPD	5 7 CPU	6 8 Q_2	7 9 Q_3	8 10 GND	9 12 P_3	10 13 P_2	11 14 PE	12 15 TCU	13 17 TCD	14 18 MR	15 19 P_0	16 20 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
7 T_C = +25°C	Functional 4/	3014	52	A	L	L	A	A	L	L	GND	A	A	A	H	H	A	A	B/	All outputs			
			53		H	H			H	H				B			B						
			54											A									
			55	B								B	B	A				B					
			56		L	L			L	L				B									
			57											A									
			58					B															
			59		L	H		A															
			60		H			B															
			61			L		A															
			62			H		B															
			63			L		A															
			64			H		B															
			65		L	L		A	H														
			66			H		B															
			67			L		A															
			68			H		B															
			69		M	L		A															
			70			H		B															
			71			H		B															
			72			L		B															
			73		L	L		A	L	H													
			74			H		B															
			75			L		A															
			76			H		B															
			77			L		A							L	H							
			78	A	L	H		B				A											
			79					A		H				B									
			80					B						A									
			81			H		A															
			82			L		B							L	H							
			83					A	H	L													
			84	B		L		B		H			A										
			85					A		L				B									
			86			H		B						A									
			87			H		A		L													
			88			L		B							L	H							
			89					A		L													
			90	A		M		B															
			91					A		L				B									
			92			H		B						A									
			93			L		A		L													
			94			H		B							L	H							
			95			L		A	L	L													
			96			H		B									A						
			97			L		A	H	H				B									
			98			H		B						A									
			99			L		A															
			100			H		B															
			101			H		A															
			102			L		B															
			103			L		A															
			104			H		B															
			105		M	L		A		L													
			106			H		B															
			107			L		A															
			108			H		B															
			109		L	L		A		H													
			110			H		B															
			111			L		A															
			112			H		B															
			113		M	L		A		L													
			114			H		B															
			115			L		A															
			116			H		B															
			117		L	L		A	H								B						
			118			H		B															
			119			L		A		L													

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 3 / 2	Test no.	1 P1	2	3 O11	4 Q0	5 CPD	6 CPU	7 Q2	8 Q3	9 GND	10 P3	11 P2	12 PE	13 TCU	14 TCD	15 MR	16 P0	17 VCC	Measured terminal	Test Limits Min	Max	Unit
7 T_C = +25°C	Functional 4/	3014		120	A		L	L	B	A	H	L	GND	A	A	H		H	B	A	B/	All outputs			
				121	H		H	H	A		L	H													
				122				L	B		H														
				123				H	A		L														
				124				L	B		H														
				125			L	H	A		L														
				126			H	L	B		H														
				127			L	H	A		L														
				128			H	L	B		H														
				129			H	L	A		L														
				130			L	H	B		H														
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																								
9 T_C = +25°C	t_{PHL1} 5/	3003 Fig. 5		131			OUT	OUT	2.7 V	IN			GND		2.7 V				GND		5.0 V		90		ns
				132							OUT														
				133								OUT													
				134					IN	2.7 V															
				135																					
				136			OUT																		
				137							OUT														
				138								OUT													
	t_{PLH1}			139	10.0 V		OUT	OUT	2.7 V	IN				0.0 V	10.0 V	2/			10.0 V	10.0 V			4.0	8.5	
				140	10.0 V																				
				141							OUT														
				142								OUT													
				143					IN	2.7 V					10.0 V					10.0 V					
				144	10.0 V		OUT																		
				145							OUT			0.0 V	10.0 V										
				146								OUT													
	t_{PHL1}			147			OUT	OUT	2.7 V	IN													4.5	12.5	
				148							OUT														
				149								OUT													
				150					IN	2.7 V															
				151	10.0 V		OUT							0.0 V	10.0 V				10.0 V						
				152							OUT				10.0 V										
				153	10.0 V							OUT													
				154	10.0 V										10.0 V										
	t_{PLH2}			155					2.7 V	IN							OUT								
	t_{PHL2}			156					2.7 V	IN							OUT								
	t_{PLH3}			157	10.0 V		OUT		10.0 V	10.0 V						IN			2/	12.7 V			5.0	111.0	
				158	12.7 V									0.0 V	10.0 V					0.0 V					
				159	10.0 V						OUT				12.7 V										
				160								OUT		12.7 V	10.0 V										
	t_{PHL3}			161			OUT	OUT												10.0 V			5.5	113.0	
				162							OUT			0.0 V	10.0 V										
				163								OUT													
				164																					
	t_{PLH4}			165					IN	2.7 V						2/		OUT					4.0	9.0	
	t_{PHL4}			166					IN	2.7 V						2/		OUT					3.5	8.0	

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.9 V, or open)

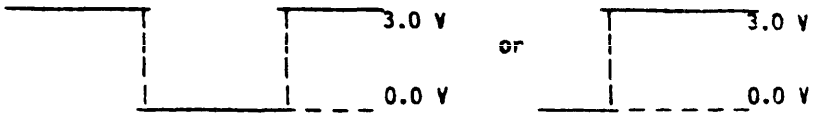
Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				P_1	Q_1	Q_0	CPD	CPU	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	$\overline{TCU}$	$\overline{TCD}$	MR	P_0	V_{CC}							
9 T_C = +25°C	t_{PLH5}	3003 Fig. 5	167			OUT	0.0 V	0.0 V			GND			0.0 V			0.0 V	IN	5.0 V	P_0 to Q_0	3.0	7.0	ns
			168	IN	OUT		"	"			"			"			"		"	P_1 to Q_1	"	"	"
			169				"	"	OUT		"		IN	"			"		"	P_2 to Q_2	"	"	"
			170				"	"		OUT	"	IN		"			"		"	P_3 to Q_3	"	"	"
	t_{PHL5}		171			OUT	"	"			"			"			"	IN	"	P_0 to Q_0	6.0	14.5	"
			172	IN	OUT		"	"			"			"			"		"	P_1 to Q_1	"	"	"
			173				"	"	OUT		"		IN	"			"		"	P_2 to Q_2	"	"	"
			174				"	"		OUT	"	IN		"			"		"	P_3 to Q_3	"	"	"
	t_{PLH5}		175	0.0 V			2.7 V	"			"		0.0 V	2/	OUT		IN	"	"	MR to $\overline{TCU}$	"	13.5	"
	t_{PHL6}		176	"			0.0 V	2.7 V			"		"	2/		OUT	IN	"	"	MR to $\overline{TCD}$	7.0	14.5	"
	t_{PL47}		177	"			2.7 V	0.0 V			"		"	IN	OUT		0.0 V	0.0 V	"	$\overline{PL}$ to $\overline{TCU}$	"	15.5	"
	t_{PLH8}		178	"			0.0 V	2.7 V			"		"	"		OUT	2/		"	$\overline{PL}$ to $\overline{TCD}$	"	15.5	"
	t_{PHL7}		179	"			2.7 V	0.0 V			"		"	"	OUT		2/		"	$\overline{PL}$ to $\overline{TCU}$	"	14.5	"
	t_{PHL8}		180	"			0.0 V	2.7 V			"	0.0 V	"	"		OUT	0.0 V	0.0 V	"	$\overline{PL}$ to $\overline{TCD}$	"	"	"
	t_{PL49}		181	"			2.7 V	0.0 V			"		"	0.0 V	"		"	IN	"	P_0 to $\overline{TCU}$	"	"	"
			182	"			"	"			"	IN	"	"			"		"	P_3 to $\overline{TCU}$	"	"	"
	t_{PHL9}		183	"			"	"			"		"	"	"		"	IN	"	P_0 to $\overline{TCU}$	6.5	14.0	"
			184	"			"	"			"	IN	"	"			"		"	P_3 to $\overline{TCU}$	"	"	"
	t_{PLH10}		185	0.0 V			0.0 V	2.7 V			"	0.0 V	0.0 V	"		OUT	"	IN	"	P_0 to $\overline{TCD}$	7.0	14.5	"
			186	IN			"	"			"	"	0.0 V	"		"	"	0.0 V	"	P_1 to $\overline{TCD}$	"	"	"
			187	0.0 V			"	"			"	"	IN	"		"	"	"	"	P_2 to $\overline{TCD}$	"	"	"
			188	"			"	"			"	IN	0.0 V	"		"	"	"	"	P_3 to $\overline{TCD}$	"	"	"
	t_{PHL10}		189	"			"	"			"	0.0 V	"	"		"	"	IN	"	P_0 to $\overline{TCD}$	6.5	14.0	"
			190	IN			"	"			"	"	"	"		"	"	0.0 V	"	P_1 to $\overline{TCD}$	"	"	"
			191	0.0 V			"	"			"	"	IN	"		"	"	"	"	P_2 to $\overline{TCD}$	"	"	"
			192	"			"	"			"	IN	0.0 V	"		"	"	"	"	P_3 to $\overline{TCD}$	"	"	"
	t_{PHL11}		193	"		OUT	"	0.0 V			"	0.0 V	"	2/			IN		"	MR to Q_0	"	14.5	"
			194		OUT		"	"			"	"	"	"			"	0.0 V	"	MR to Q_1	"	"	"
			195	0.0 V			"	"	OUT		"	"	"	"			"	"	"	MR to Q_2	"	"	"
			196	0.0 V			"	"		OUT	"		0.0 V	"			"	"	"	MR to Q_3	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits for table I.																										
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

See footnotes at end of table.

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1/ For case 2, pins not referenced are NC.

2/ Apply one pulse prior to measurement.



3/

Parameter	Device	Circuit		
		A	B	C
I_{IL1}	ALL	-.25/-0.6	-.03/-0.6	
I_{IL2}	01, 02	-.50/-1.2	-.50/-1.2	
I_{IL3}	03, 04	-.75/-1.8	-.75/-1.8	

4/ H = 2.5 V, L = 0.5 V, A = 3.0 V minimum; B = 0.0 V or GND.

5/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

6/ For types 01 and 02, set outputs to 9th count (P_0 = 1, P_3 = 1, P_1 and P_2 = 0).

7/ For types 01 and 02 increment such that measurement of the specified output can occur on the next applied CP.

8/ Perform entire functional testing at 4.5 V_{CC} and entire functional testing again at 5.5 V_{CC}.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirments for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5005 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for products assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F160A
02	54F162A
03	54F190
04	54F192

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6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit				
	A	B	C	D	E
	Fairchild Semiconductor	Motorola	Signetics		
01	X				
02	X				
03					
04	X				

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/344	MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, DECADE COUNTERS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify) _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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